Structure from Observation:

Probability, Reconstruction, and Finite-Sample Certification

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Abstract

This monograph develops a unified programme showing that probability, dynamics, reconstruction, and their finite-sample signatures are not assumptions but consequences of coherent structured observation.

Part I answers the question: when does observational coherence determine probability? A coherent family of observations on a directed system of observable algebras extends to a unique σ-additive probability measure exactly when it satisfies collective exhaustion (CE) — the condition ruling out persistent mass on globally vanishing events. Two independent routes establish this: Carathéodory extension on the realization space, and compactness of the Stone space. CE is shown to be irreducible: no first-order condition on the query system implies it.

Part II answers the question: given probability, how do dynamics and reconstruction arise from observation? Making prediction precise forces the introduction of predictive kernels; their temporal composition yields a Markov semigroup and Koopman–Perron duality, derived rather than assumed. Specialising the minimal predictive state map to a measure-preserving system yields the delay map, and the reconstruction question is internal to the same framework: whether the predictive state is faithful. The answer is a three-way equivalence, with the density bridge lemma as the key. When reconstruction holds, the Stone space of Part I is identified with the state space itself.

Part III answers the question: given reconstruction as the asymptotic notion, what can finite data certify? Three theorems — the Algebra Theorem, the Dynamics Theorem, and the Conjunction Theorem — provide computable witnesses for reconstruction, optimal convergence rates, and the result that for deterministic systems all three witnesses certify the same object.

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Part I

When Does Observational Coherence Determine Probability?

Abstract

We prove that a coherent family of finitely additive charges satisfying *collective exhaustion* (CE) determines a unique probability measure on the observable σ-algebra. The setting is a directed system of measurable outcome spaces with surjective refinement maps and a compatible family of charges on the cylinder algebra.

A direct *Carathéodory construction* assumes compatible σ-additive marginals and builds the global measure directly without topological assumptions.

The necessary and sufficient condition for a compatible family of finitely additive charges to extend to a σ-additive measure on the observable σ-algebra is CE: no level may assign persistent mass to events the system as a whole sees as empty. CE is not derivable from any structural condition on the query system; the obstruction is located by the finite-cofinite counterexample and a model-theoretic argument via Łoś's theorem.

A *Stone construction* starts from finite additivity alone. Compactness of the Stone space supplies σ-additivity on the compact completion; CE then appears as the support condition ensuring the resulting measure descends to the realised instantiation space. The two constructions differ by where the missing σ-additive structure is supplied — internally by countable reach in the directed system, or externally by the Stone completion — and whenever both constructions apply, they yield the same observable measure.

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1 Introduction

Observable distinctions are classically typified by Boolean algebras. Their study crucially underpins modern measure theory and its probabilistic extensions. In contrast, the directed nature of observation, wherein finer levels of resolution refine coarser ones, has been relatively misprized toward such ends. In the foundational treatments of Kolmogorov [14] and Choksi [5], the global σ-algebra emerges as the projective limit of compatible measure spaces. Directed structure plays an organisational role: it orders the marginals and licenses the assembly, but is not itself interrogated as a source of probabilistic constraint.

We facilitate interrogation by a directed system of Boolean algebras connected by surjective refinement maps. Valuations enter as compatible families of finitely additive charges on the level algebras: charges at different levels agree on events seen from both. Structural coherence and valuation compatibility together constitute the finitary data.

Under such treatment Kolmogorov's question resurfaces without dependence on the ontological status of a global state space. In the language of *queries*: what further condition on valuations is required for finitary data to support a genuine probability measure on the observable σ-algebra?

The condition is *collective exhaustion* (CE): a compatible family of finitely additive charges extends to a σ-additive measure if and only if, whenever a decreasing sequence of events vanishes globally, there exists a level at which the sequence's mass converges to zero. CE rules out purely finitely additive charges, which assign persistent mass to sequences that vanish in the limit. It is the exact admissibility condition separating observational coherence from genuine probability: a coherent family of charges deserves to be treated as a probability system precisely when it satisfies CE. A family satisfying CE determines a unique probability measure on the observable σ-algebra; no first-order structural condition can substitute for it.

2 Observational structure and compatible charges

2.1 Observational structure

Rather than a single σ-algebra of events, we work with a directed system of Boolean algebras representing observable distinctions at increasing resolution; the global σ-algebra arises as their completion.

Definition 2.1 (Observational structure). An *observational structure* consists of:

- a nonempty partially ordered set $(\iota, \leq)$, the *index set*;
- for each $i \in \iota$, a measurable space $(\mathsf{O}_i, \mathcal{O}_i)$, the *outcome space at level* i;
- for each $i \leq j$ in ι, a measurable surjection $\pi_{ij} : \mathsf{O}_j \to \mathsf{O}_i$, the *refinement map*,

satisfying $\pi_{ik} = \pi_{ij} \circ \pi_{jk}$ for all $i \leq j \leq k$.

Definition 2.2 (Instantiation; query system). An *instantiation* of an observational structure $(\iota, \leq, \{(\mathsf{O}_i, \mathcal{O}_i)\}, \{\pi_{ij}\})$ is a set Ω together with maps $\mathrm{eval}_i : \Omega \to \mathsf{O}_i$ for each $i \in \iota$, the *evaluation maps*, satisfying the *coherence condition* $\pi_{ij} \circ \mathrm{eval}_j = \mathrm{eval}_i$ for all $i \leq j$. A *query system* is an observational structure together with an instantiation.

2.2 Canonical instantiation and observables

Every observational structure admits a canonical instantiation: the space of all coherent assignments of outcomes across levels. This identifies Ω with the set of all globally consistent observation patterns. Explicitly, the canonical instantiation is the projective limit

$$\Omega = \varprojlim_{i \in \iota} \mathsf{O}_i = \Big\{ \omega \in \prod_{i \in \iota} \mathsf{O}_i \;\Big|\; \pi_{ij}(\omega_j) = \omega_i \text{ whenever } i \leq j \Big\},$$

with $\mathrm{eval}_i(\omega) = \omega_i$ and $\omega \in \Omega$ generic.

Example 2.3 (Bernoulli observations). Let ι be the set of finite subsets of $\mathbb{N}$, ordered by inclusion. For $i \in \iota$, set $\mathsf{O}_i = \{0,1\}^{|i|}$ with its discrete σ-algebra $\mathcal{O}_i$, and for $i \subseteq j$ let $\pi_{ij} : \mathsf{O}_j \to \mathsf{O}_i$ be the coordinate projection. An instantiation assigns to each ω a consistent family of finite binary observations; the canonical instantiation identifies Ω with $\{0,1\}^{\mathbb{N}}$. Cylinder sets correspond to fixing finitely many coordinates, and a compatible family $\{\nu_i\}$ is precisely the system of finite-dimensional distributions of a Bernoulli process.

Definition 2.4 (Common coarsenings). The index set $(\iota, \leq)$ admits *common coarsenings* if every pair $i, j \in \iota$ has a lower bound: there exists $k \leq i$ and $k \leq j$.

Definition 2.5 (Cylinder sets). For $i \in \iota$ and $A \in \mathcal{O}_i$, the *level-i cylinder with base A* is

$$\mathrm{Cyl}(i, A) \;=\; \{\omega \in \Omega : \mathrm{eval}_i(\omega) \in A\}.$$

The *cylinder family* is $\mathcal{C} = \{\mathrm{Cyl}(i, A) : i \in \iota,\ A \in \mathcal{O}_i\}$, and the *observable σ-algebra* is $\sigma(\mathcal{C})$.

Lemma 2.6 (Cylinder π-system). *Assume $(\iota, \leq)$ admits common coarsenings. Then $\mathcal{C}$ is a π-system: it is closed under finite intersections.*

Proof. The coherence condition gives $\mathrm{Cyl}(i, A) = \mathrm{Cyl}(j, \pi_{ij}^{-1}(A))$ for $i \le j$: the same subset of Ω is a cylinder at any finer level. Given $\mathrm{Cyl}(i, A)$ and $\mathrm{Cyl}(j, B)$, let k be a common lower bound. Then

$$\mathrm{Cyl}(i, A) \cap \mathrm{Cyl}(j, B) = \mathrm{Cyl}(k,\ \pi_{ki}^{-1}(A) \cap \pi_{kj}^{-1}(B)),$$

which is again a cylinder. □

For each $i \in \iota$, let B_i denote the Boolean algebra of level-i cylinders $\{\mathrm{Cyl}(i, A) : A \in \mathcal{O}_i\}$; then $\mathcal{C} = \bigcup_{i\in\iota} B_i$. For $i \le j$, define the *connecting map*

$$\varphi_{ij} : B_i \to B_j, \qquad \varphi_{ij}(\mathrm{Cyl}(i, A)) = \mathrm{Cyl}(j, \pi_{ij}^{-1}(A)).$$

This is a well-defined Boolean algebra homomorphism. The family $\{B_i, \varphi_{ij}\}$ is a directed system of Boolean algebras.

2.3 Compatible charges

Definition 2.7 (Compatible family)**.** A *charge* on a Boolean algebra B is a finitely additive function $\mu : B \to [0, \infty)$ with $\mu(\emptyset) = 0$. A family of charges $\{\mu_i\}_{i\in\iota}$ with μ_i on B_i is *compatible* if

$$\mu_j(\varphi_{ij}(E)) = \mu_i(E) \qquad \text{for all } i \le j \text{ and all } E \in B_i.$$

A compatible family also determines a charge on $\mathcal{C}$ by $\mu(\mathrm{Cyl}(i, A)) := \mu_i(\mathrm{Cyl}(i, A))$; compatibility ensures this is well-defined.

Both constructions require the following condition on the query system.

Definition 2.8 (Surjective Evaluation)**.** The query system satisfies *Surjective Evaluation* if $\mathrm{eval}_i : \Omega \to \mathsf{O}_i$ is surjective for every $i \in \iota$.

Surjective evaluation ensures every outcome at every level is realised by some $\omega \in \Omega$.

Lemma 2.9 (Surjectivity and injectivity from Surjective Evaluation)**.** *If the query system satisfies Surjective Evaluation, then for every $i \le j$:*

(i) the refinement map $\pi_{ij} : \mathsf{O}_j \to \mathsf{O}_i$ is surjective;

(ii) the connecting map $\varphi_{ij} : B_i \to B_j$, defined by $\varphi_{ij}(\mathrm{Cyl}(i, A)) = \mathrm{Cyl}(j, \pi_{ij}^{-1}(A))$, is an injective Boolean algebra homomorphism.

Proof. For (i): given $y \in \mathsf{O}_i$, surjectivity of eval_i yields $\omega \in \Omega$ with $\mathrm{eval}_i(\omega) = y$. Then $\pi_{ij}(\mathrm{eval}_j(\omega)) = \mathrm{eval}_i(\omega) = y$ by the coherence condition, so π_{ij} is surjective.

For (ii): φ_{ij} is a Boolean homomorphism by construction. If $\varphi_{ij}(\mathrm{Cyl}(i, A)) = \varphi_{ij}(\mathrm{Cyl}(i, A'))$, then $\pi_{ij}^{-1}(A) = \pi_{ij}^{-1}(A')$ in O_j. Applying π_{ij} and using surjectivity gives $A = A'$, so φ_{ij} is injective. □

3 Direct extension from compatible probability marginals

Given compatible σ-additive probability marginals, Carathéodory's theorem yields a global measure directly on the instantiation space. The input is a query system satisfying surjective evaluation, sequential common refinements, and common coarsenings, together with a compatible family of σ-additive probability marginals.

3.1 Observational determination

Proposition 3.1 (Observational determination)**.** *Let the query system admit common coarsenings and let P, P' be probability measures on $(\Omega, \sigma(\mathcal{C}))$. If $(\mathrm{eval}_i)_\# P = (\mathrm{eval}_i)_\# P'$ for all $i \in \iota$, then $P = P'$.*

Proof. The equality of pushforward laws implies P and P' agree on every cylinder event. By Lemma 2.6 the cylinder family $\mathcal{C}$ is a π-system generating $\sigma(\mathcal{C})$. The π–λ theorem gives $P = P'$. □

3.2 Cylinder content

Definition 3.2 (Common refinements)**.** The index set $(\iota, \leq)$ admits *common refinements* if every pair $i, j \in \iota$ has an upper bound: there exists $k \geq i$ and $k \geq j$.

The cylinder family may represent the same event at multiple levels: $\mathrm{Cyl}(i, A)$ and $\mathrm{Cyl}(j, B)$ coincide in Ω whenever A and B are preimages of a common set at a finer level. Assigning a numerical value to such an event is well-defined only if the two representations agree — and verifying this requires passing to a common refinement of i and j.

Lemma 3.3 (Finite observational content)**.** *Assume the query system admits common coarsenings and common refinements and satisfies Surjective Evaluation, and let $\{\nu_i\}_{i\in\iota}$ be a family with each $\nu_i \in \mathcal{P}(\mathsf{O}_i)$ compatible under the refinement maps, i.e. $(\pi_{ij})_\# \nu_j = \nu_i$ for all $i \leq j$. Then the assignment*

$$\mu_0(\mathrm{Cyl}(i, A)) := \nu_i(A), \qquad A \in \mathcal{O}_i,$$

defines a well-defined finitely additive content on $\mathcal{C}$.

Proof. Well-definedness. Suppose $\mathrm{Cyl}(i, A) = \mathrm{Cyl}(j, B)$. Use common refinements to find k with $k \geq i$ and $k \geq j$. The constraint sets in O_k are $\pi_{ik}^{-1}(A)$ and $\pi_{jk}^{-1}(B)$. The cylinder equality and surjective evaluation (surjectivity of eval_k) imply these coincide in O_k. Compatibility then gives $\nu_i(A) = \nu_k(\pi_{ik}^{-1}(A)) = \nu_k(\pi_{jk}^{-1}(B)) = \nu_j(B)$.

Finite additivity. For disjoint cylinders at possibly different levels, compress to a common refinement k and use additivity of ν_k. □

3.3 Carathéodory extension theorem

Definition 3.4 (Sequential common refinements)**.** The index set $(\iota, \leq)$ admits *sequential common refinements* if every countable family $\{i_n\}_{n\geq 1} \subset \iota$ has an upper bound in ι. Sequential common refinements imply common refinements.

Finite additivity of the cylinder content is not sufficient for Carathéodory's theorem: one needs σ-subadditivity, which requires compressing a countable cylinder cover $\{\mathrm{Cyl}(i_n, A_n)\}$ to a single level. Common refinements handle finite families; sequential common refinements extend this to countable ones, supplying the index-set reach the proof requires.

Theorem 3.5 (Carathéodory observational extension)**.** *Suppose:*

(i) $(\iota, \leq)$ admits common coarsenings and sequential common refinements;

(ii) the query system satisfies Surjective Evaluation;

(iii) each ν_i is a probability measure on $(\mathsf{O}_i, \mathcal{O}_i)$, and the family is compatible under the refinement maps: $(\pi_{ij})_\# \nu_j = \nu_i$ for all $i \leq j$.

Then there exists a unique probability measure P on $(\Omega, \sigma(\mathcal{C}))$ such that $(\mathrm{eval}_i)_\# P = \nu_i$ for every $i \in \iota$.

Proof. Step 1: Finite content. Lemma 3.3 gives a well-defined finitely additive content μ_0 on $\mathcal{C}$.

Step 2: σ-subadditivity. Let $\mathrm{Cyl}(i, B) \subseteq \bigcup_{n\geq 1} \mathrm{Cyl}(i_n, A_n)$. Apply sequential common refinements to $\{i, i_1, i_2, \ldots\}$ to find a single index k with $k \geq i$ and $k \geq i_n$ for all n. Compress each cylinder to level k:

$$\mathrm{Cyl}(i, B) = \mathrm{Cyl}(k, E), \qquad \mathrm{Cyl}(i_n, A_n) = \mathrm{Cyl}(k, E_n),$$

where $E = \pi_{ik}^{-1}(B)$ and $E_n = \pi_{i_n k}^{-1}(A_n)$. Surjective evaluation and the cylinder inclusion in Ω imply $E \subseteq \bigcup_n E_n$ in O_k. Therefore

$$\mu_0(\mathrm{Cyl}(i, B)) = \nu_k(E) \leq \sum_{n=1}^{\infty} \nu_k(E_n) = \sum_{n=1}^{\infty} \mu_0(\mathrm{Cyl}(i_n, A_n)),$$

using σ-subadditivity of the single measure ν_k.

Step 3: Carathéodory extension. The family $\mathcal{C}$ is a set semiring generating $\sigma(\mathcal{C})$. It is closed under finite intersections by Lemma 2.6. Moreover, if $E = \mathrm{Cyl}(i, A)$ and $F = \mathrm{Cyl}(j, B)$, choose k with $k \geq i, j$; then

$$E = \mathrm{Cyl}\big(k,\, \pi_{ik}^{-1}(A)\big), \qquad F = \mathrm{Cyl}\big(k,\, \pi_{jk}^{-1}(B)\big),$$

so

$$E \setminus F = \mathrm{Cyl}\big(k,\, \pi_{ik}^{-1}(A) \setminus \pi_{jk}^{-1}(B)\big) \in \mathcal{C}.$$

Thus $\mathcal{C}$ is closed under relative differences, confirming the semiring property. Since μ_0 is a finitely additive content on $\mathcal{C}$ that is σ-subadditive for countable covers by cylinder sets (Step 2), the semiring form of Carathéodory's extension theorem yields a measure P on $(\Omega, \sigma(\mathcal{C}))$ extending μ_0 (Bogachev [3], Vol. I, Corollary 1.11.9).

Step 4: Marginal recovery and normalization. For any $A \in \mathcal{O}_i$, $P(\mathrm{Cyl}(i, A)) = \mu_0(\mathrm{Cyl}(i, A)) = \nu_i(A)$, so $(\mathrm{eval}_i)_\# P = \nu_i$. Setting $A = \mathsf{O}_i$ gives $P(\Omega) = 1$.

Uniqueness. Two probability measures with the same evaluation marginals agree on $\mathcal{C}$, hence on $\sigma(\mathcal{C})$ by Proposition 3.1. ☐

Remark 3.6 (Relation to Kolmogorov's extension theorem)**.** Kolmogorov's theorem [14] constructs a probability measure on S^T from consistent finite-dimensional distributions. Theorem 3.5 plays the same role for query systems: the instantiation $\Omega = \varprojlim \mathsf{O}_i$ replaces the product S^T, compatible marginals replace consistent finite-dimensional distributions, and sequential common refinements replace the countable-cofinality condition. The key conceptual difference is that no topology is imposed on the outcome spaces: the measure P is built directly on the measurable structure of Ω by Carathéodory, without any appeal to tightness or compactness.

4 Collective Exhaustion

Compatible σ-additive marginals extend directly to a global observable measure. The obstruction to passing from finite additivity to σ-additivity is the possibility that mass persists along a decreasing sequence of events whose intersection is empty.

Example 4.1 (Finite–cofinite obstruction)**.** Let $\mathbb{N}$ carry the finite–cofinite algebra $\mathcal{E}$ where $A \in \mathcal{E}$ if A is finite or A^c is finite. Define a normalized charge by $\mu_0(A) = 0$ if A is finite and $\mu_0(A) = 1$ if A is cofinite.

Disjoint sets in $\mathcal{E}$ are either both finite or one finite and one cofinite (two disjoint cofinite sets cannot exist), so finite additivity holds.

The sequence $E_n = \{n, n+1, \ldots\}$ satisfies $E_n \downarrow \varnothing$, yet each E_n is cofinite, so $\mu_0(E_n) = 1$ for all n: the events shrink to nothing but the mass does not drain away. No σ-additive extension exists.

The admissibility condition that rules out this failure is:

Definition 4.2 (Collective Exhaustion)**.** A charge μ on $\mathcal{C}$ (the common extension of a compatible family $\{\mu_i\}$) satisfies *collective exhaustion* (CE) if: whenever $E_1 \supseteq E_2 \supseteq \cdots$ in $\mathcal{C}$ and $\bigcap_n E_n = \emptyset$, we have $\mu(E_n) \to 0$.

CE is a valuation-layer condition: it constrains how charges behave, not the Boolean-algebraic index structure.

We work first at the level of an abstract directed system of Boolean algebras, writing $\mathcal{E}_i$ for the algebra at level i and ℓ_i for its charge; the query-system interpretation returns in §5.

4.1 Single-algebra extension

Definition 4.3 (Boolean charge space)**.** A *Boolean charge space* is a pair $(\mathcal{E}, \ell)$ where $\mathcal{E}$ is a Boolean algebra of subsets of a set O and $\ell : \mathcal{E} \to [0, 1]$ is a normalized finitely-additive valuation. No σ-algebra and no topology are assumed on O.

By Stone's representation theorem [22], $\mathcal{E}$ is isomorphic to the clopen algebra of a compact totally-disconnected Hausdorff space $S(\mathcal{E})$, and $\sigma(\mathcal{E})$ is the Baire σ-algebra of $S(\mathcal{E})$. The sets in $\sigma(\mathcal{E}) \setminus \mathcal{E}$ are Baire events that countable operations can reach but no single observable distinction can name.

Lemma 4.4 (Extension criterion)**.** *A normalized finitely-additive $\ell : \mathcal{E} \to [0, 1]$ extends to a σ-additive measure on $\sigma(\mathcal{E})$ if and only if ℓ is continuous at $\emptyset$: for every decreasing sequence $E_1 \supseteq E_2 \supseteq \cdots$ in $\mathcal{E}$ with $\bigcap_n E_n = \emptyset$ in $\sigma(\mathcal{E})$, we have $\ell(E_n) \to 0$. The extension is then unique.*

Proof. Standard; see Halmos [11] or Fremlin [8] Vol. 1. Continuity at $\emptyset$ is used in the Carathéodory extension argument to pass from finite additivity to σ-additivity on $\sigma(\mathcal{E})$. $\square$

The condition in Lemma 4.4 is not verifiable within $\mathcal{E}$ alone: confirming $\bigcap_n E_n = \emptyset$ requires access to $\sigma(\mathcal{E})$. In a directed system, one might hope a finer level $j \geq i$ could witness the emptiness level i cannot see.

4.2 Nontrivial refinement

Definition 4.5 (Nontrivial refinement)**.** Let $\pi : O_j \to O_i$ be a surjective refinement map, inducing the Boolean algebra homomorphism $\pi^{-1} : \mathcal{E}_i \to \mathcal{E}_j$. The refinement is *nontrivial* if there exists $A \in \mathcal{E}_j$ with $A \notin \pi^{-1}(\mathcal{E}_i)$.

Proposition 4.6 (Nontrivial refinement introduces new events)**.** *If $j \geq i$ is a nontrivial refinement, then $\pi^{-1}(\mathcal{E}_i) \subsetneq \mathcal{E}_j$. Equivalently, a level i at which $\mathcal{E}_j = \pi^{-1}(\mathcal{E}_i)$ for every $j \geq i$ is one above which the refinement order is algebraically trivial: no finer level adds new discriminative power.*

Proof. Nontriviality asserts the existence of $A \in \mathcal{E}_j$ with $A \notin \pi^{-1}(\mathcal{E}_i)$, directly giving the strict inclusion. The contrapositive is the stated equivalence. $\square$

Lemma 4.7 (Strict enlargement under σ-closure)**.** *Let $\mathcal{E}$ be a Boolean algebra of subsets of a countably infinite set O that is not a σ-algebra. Then $\sigma(\mathcal{E}) \setminus \mathcal{E} \neq \emptyset$.*

Proof. If $\mathcal{E}$ is not a σ-algebra, there exists a countable family $\{A_n\} \subset \mathcal{E}$ with $\bigcup_n A_n \notin \mathcal{E}$. By definition $\bigcup_n A_n \in \sigma(\mathcal{E})$, so it lies in $\sigma(\mathcal{E}) \setminus \mathcal{E}$. $\square$

4.3 CE characterises σ-additivity

The directed-system version of CE and the main equivalence are as follows.

Definition 4.8 (Collectively exhaustive family)**.** A compatible family $\{\ell_i\}$ of normalized finitely-additive charges on a directed system is *collectively exhaustive* if: for every $i \in \mathcal{I}$ and every decreasing sequence $E_1 \supseteq E_2 \supseteq \cdots$ in $\mathcal{E}_i$ with $\bigcap_n E_n = \emptyset$, there exists $j \geq i$ such that $\ell_j(\pi_{ij}^{-1}(E_n)) \to 0$.

$\pi_{ij} : O_j \to O_i$ is the refinement map (j finer than i) and $\pi_{ij}^{-1}(E_n) \subseteq O_j$ is the preimage of E_n at the finer level. No level may assign persistent mass to events the system as a whole sees as empty.

We identify the minimal directed system in which the index-layer hypotheses hold while the finite-cofinite charge still fails σ-additive extension.

Proposition 4.9 (Index-layer conditions do not force σ-additivity)**.** *There exists a compatible family of normalized finitely-additive charges on a sequentially upper-directed directed system that fails σ-additive extension at every level.*

Proof. We take the simplest sequentially upper-directed index set: adjoin a single terminal element ω to $\mathbb{N}$, setting $n \leq \omega$ for all $n \in \mathbb{N}$. Take all outcome spaces to be $\mathbb{N}$, all event algebras to be the finite-cofinite algebra $\mathcal{E}$, all refinement maps to be the identity, and each ℓ_i to be the finite-cofinite charge ℓ of Example 4.1. Compatibility holds trivially. The system satisfies every structural condition. By Example 4.1, ℓ is not σ-additive and admits no σ-additive extension; hence every level fails σ-additive extension. $\square$

Thus the obstruction reappears: finite additivity at j does not force continuity at $\emptyset$.

Theorem 4.10 (CE Characterisation)**.** *Let $\{\ell_i\}$ be a normalized compatible family of finitely-additive charges on a directed system indexed by $\mathcal{I}$. The following are equivalent:*

(i) The family is collectively exhaustive.

(ii) ℓ_i extends uniquely to a σ-additive measure on $\sigma(\mathcal{E}_i)$ for every $i \in \mathcal{I}$.

Proof. (i) $\Rightarrow$ (ii). Fix $i \in \mathcal{I}$ and a decreasing sequence $E_n \searrow \emptyset$ in $\mathcal{E}_i$. By CE, find $j \geq i$ with $\ell_j(\pi_{ij}^{-1}(E_n)) \to 0$. By compatibility, $\ell_i(E_n) = \ell_j(\pi_{ij}^{-1}(E_n)) \to 0$. So ℓ_i is continuous at $\emptyset$. Lemma 4.4 gives the unique σ-additive extension.

(ii) $\Rightarrow$ (i). Suppose ℓ_i extends to μ_i for every i. Let $E_n \searrow \emptyset$ in $\mathcal{E}_i$. Then $\ell_i(E_n) = \mu_i(E_n) \to \mu_i(\emptyset) = 0$ by σ-additivity of μ_i and normalization. Take $j = i$. $\square$

The question "do the structural conditions on a directed system force σ-additivity?" is ill-posed: it conflates conditions on the index structure with conditions on the charges. Proposition 4.9 separates them; Theorem 4.10 identifies the exact valuation-layer condition. Sequential upper-directedness enters only when assembling per-level extensions into a global measure (§3).

4.4 CE is non-derivable

The Yosida–Hewitt decomposition [24] identifies the obstruction Proposition 4.9 has exposed. Every finitely additive probability decomposes uniquely as $\ell = \ell_c + \ell_p$, where ℓ_c is σ-additive and ℓ_p is *purely finitely additive.* The finite-cofinite content has $\ell_p = \ell$. By Theorem 4.10, CE holds precisely when $\ell_p = 0$. The obstruction is therefore not a feature of any particular structural hypothesis: it is the persistence of a purely finitely additive component that no first-order structural condition can eliminate.

Proposition 4.11 (CE non-derivability)**.** *Collective exhaustion is not implied by any structural condition on a query system.*

(i) (Particular case.) *Sequential upper-directedness, compatibility, and normalization do not imply CE.*

(ii) (Universal case.) *No condition Φ expressible in the first-order language of query systems implies CE.*

Proof. The particular case is Proposition 4.9: the finite-cofinite charge satisfies all named structural hypotheses and fails CE.

For the universal case, we proceed in three steps.

Step 1. A first-order condition on query systems is preserved under ultraproducts by Łoś's theorem [16]: if Φ holds in every member of a family of query systems, it holds in their ultraproduct.

Step 2. Since the finite-cofinite construction works on any countably infinite set, we may realise it on $\mathbb{Q}$. Fix an enumeration $q : \mathbb{N} \to \mathbb{Q}$, let $\mathcal{U}$ be a nonprincipal ultrafilter on $\mathbb{N}$ extending the cofinite filter, and let δ_n be the point mass at $q(n) \in \mathbb{Q}$. Each δ_n is a normalized σ-additive charge satisfying every structural condition including CE. The ultraproduct $\prod_{\mathcal{U}} \delta_n$ assigns to $E \subseteq \mathbb{Q}$ the value $\lim_{\mathcal{U}} \mathbf{1}_{q(n)\in E}$.

If E is finite then $q(n) \in E$ holds for only finitely many n, so the ultralimit is 0. If E is cofinite then $q(n) \in E$ holds for cofinitely many n, so the ultralimit is 1, since $\mathcal{U}$ extends the cofinite filter. The ultraproduct is therefore exactly the finite-cofinite charge ℓ of Example 4.1 on $\mathbb{Q}$.

Step 3. By Step 1, any universally valid first-order condition Φ holds in the ultraproduct whenever it holds in each δ_n. But by Step 2 the ultraproduct is exactly ℓ, and ℓ fails CE. Therefore no first-order structural condition can imply CE. □

The ultrafilter-based charge is the canonical witness: it passes every finite consistency test yet assigns unit mass to events whose infinite intersection is empty.

4.5 Finite coherence does not determine probability

The obstruction in Proposition 4.11 is not an artifact of the query-system language. In the more primitive setting of Boolean algebras with a normalized finitely additive charge, the same conclusion holds: no first-order theory in that language characterizes those algebras whose charge extends to a σ-additive measure. Equivalently, countable additivity is not first-order axiomatizable among finitely additive probability Boolean algebras [17]. The probability logic community has treated this as background knowledge — Hoover [12] requires an infinitary rule to axiomatize σ-additive probability logic, and Fagin, Halpern, and Megiddo [7] state without proof that countable additivity cannot be expressed by any formula in the natural language — but an explicit proof does not appear in the literature. The companion note supplies one via the same ultraproduct mechanism: Dirac charges δ_n satisfy any putative characterizing theory, yet their ultraproduct is the finite-cofinite charge ℓ, which fails σ-additivity, contradicting Łoś's theorem.

Finite coherence therefore underdetermines probability in a logically precise sense. Foundational disputes about probability are not terminological; they concern which extra structure licenses the passage from finite coherence to probability, and some such structure is unavoidable. CE is the condition adopted here; the companion note [17] gives the full general proof.

Example 4.12 (Propositional probability as a query system)**.** The framework is not specific to empirical measurement. Let T be a consistent countable propositional theory with sentences $\phi_1, \phi_2, \ldots$ At level n, let

$$\mathcal{E}_n \;=\; \mathrm{Lind}(\phi_1, \ldots, \phi_n)$$

be the finite Boolean algebra of T-equivalence classes of Boolean combinations of $\phi_1, \ldots, \phi_n$. Its atoms are the maximal consistent truth-value assignments to $\phi_1, \ldots, \phi_n$ compatible with T.

For $n \leq m$, restricting an assignment on $\phi_1, \ldots, \phi_m$ to its first n sentences gives the refinement map π_{nm}. The canonical instantiation is

$$\Omega \;=\; \varprojlim_n \operatorname{atoms}(\mathcal{E}_n) \;=\; \{\text{complete consistent extensions of } T\},$$

and $\operatorname{eval}_n(\omega)$ is the restriction of a complete extension ω to $\{\phi_1, \ldots, \phi_n\}$. Surjectivity of each eval_n follows from the Lindenbaum extension lemma: every maximal consistent assignment to $\phi_1, \ldots, \phi_n$ extends to a complete consistent extension of T.

A compatible family of normalised charges $\{\nu_n\}$ on $\{\mathcal{E}_n\}$ is precisely a coherent system of finite-level probability assignments. A decreasing sequence of cylinder events $\operatorname{Cyl}(n_k, A_k)$ with $n_k \nearrow \infty$ has empty intersection in Ω if and only if the corresponding propositional conditions $\alpha_k \in \mathcal{E}_{n_k}$ are jointly unsatisfiable over T: no complete extension satisfies all of them simultaneously. CE requires that the charges drain away in this case,

$$\nu_{n_k}(\alpha_k) \;\longrightarrow\; 0,$$

ruling out persistent mass on conditions that collectively vanish. Theorem 4.10 therefore yields a unique σ-additive probability measure on the space of complete consistent extensions of T exactly when the charge family satisfies CE.

Remark 4.13. Under Stone duality, the Stone space of $\varinjlim_n \mathcal{E}_n$ is the space of possible worlds: complete consistent extensions appear as the principal ultrafilters, while non-principal ultrafilters are ideal limit points — internally coherent but not realised by any complete extension of T. By Proposition 5.7, CE is exactly the condition that no mass escapes to these ideal points. CE is thus not a condition tailored to empirical observation; it is the general admissibility principle separating coherent finite-level probability assignments from honest probability on the completed space of possible worlds.

5 Stone realization from finitely additive data

Fortunately σ-additivity can be derived. Starting only from finitely additive charges, compactness of the Stone space produces a σ-additive measure on the compact completion; CE then reappears as the support condition that returns this measure to the realised instantiation. No sequential common refinements are assumed: the σ-additivity that Carathéodory obtains by compressing countable covers within the index set is reproduced here by compactness of the Stone space, without any countable reach into ι.

5.1 Stone duality for the direct limit

Injectivity of the connecting maps (Lemma 2.9(ii)) is the key structural input for the Stone construction; we restate it here for reference.

Lemma 5.1 (Injectivity of connecting maps)**.** *If the query system satisfies Surjective Evaluation, then $\varphi_{ij} : B_i \to B_j$ is an injective Boolean algebra homomorphism for every $i \leq j$.*

Proof. Lemma 2.9(ii). □

Proposition 5.2 (Stone duality for the direct limit)**.** *The Stone space of the direct limit satisfies $\operatorname{St}\big(\bigcup_i B_i\big) \cong \varprojlim_i \operatorname{St}(B_i)$.*

Proof. By Lemma 5.1, the connecting maps φ_{ij} are injective Boolean algebra homomorphisms. Under Stone duality, injective Boolean algebra homomorphisms correspond to surjective continuous maps between Stone spaces. The Stone space functor St converts colimits in the category of Boolean algebras to limits in the category of compact Hausdorff spaces [13, II.4]. Applied to $\bigcup_i B_i$ with its injective system, this gives $\operatorname{St}\big(\bigcup_i B_i\big) \cong \varprojlim_i \operatorname{St}(B_i)$. □

5.2 The inverse limit measure

Proposition 5.3 (Inverse limit measure)**.** *Let $\{\mu_i\}$ be a compatible family of normalised charges on $\{B_i\}$. Then there exists a unique probability measure $\hat{\mu}$ on $\mathcal{B}(\varprojlim_i \mathrm{St}(B_i))$ whose pushforward along each projection $\varprojlim_i \mathrm{St}(B_i) \to \mathrm{St}(B_j)$ agrees with the Borel measure on $\mathrm{St}(B_j)$ corresponding to μ_j.*

Proof. Step 1: Single-algebra correspondence. For each i, Stone duality identifies B_i with the clopen algebra $\mathrm{Clop}(\mathrm{St}(B_i))$. A charge μ_i on B_i thus defines a finitely additive set function on the clopens of the compact Hausdorff space $\mathrm{St}(B_i)$. Since $\mathrm{St}(B_i)$ is compact and totally disconnected, the clopens form a base for the topology and a finitely additive charge on them extends uniquely to a regular Borel measure; see [4], §6, or [11], §§53–54. Call this measure $\hat{\mu}_i$ on $\mathrm{St}(B_i)$.

Step 2: Compatibility. The compatibility condition $\mu_j(\varphi_{ij}(E)) = \mu_i(E)$ translates, under Stone correspondence, to $\hat{\mu}_i$ being the pushforward of $\hat{\mu}_j$ along the bonding map $\mathrm{St}(B_j) \to \mathrm{St}(B_i)$. The family $\{\hat{\mu}_i\}$ is compatible on the inverse system $\{\mathrm{St}(B_i)\}$.

Step 3: Extension to the inverse limit. Each $\mathrm{St}(B_i)$ is compact Hausdorff. The inverse limit $\varprojlim_i \mathrm{St}(B_i)$ is compact Hausdorff. By [5, Theorem 1], a compatible family of regular Borel probability measures on a cofiltered inverse system of compact Hausdorff spaces with surjective bonding maps extends uniquely to a regular Borel probability measure on the inverse limit. This gives $\hat{\mu}$ on $\mathcal{B}(\varprojlim_i \mathrm{St}(B_i))$. □

5.3 Structural identification

The Stone space $\mathrm{St}(\mathcal{C})$ contains Ω as a subspace via the Stone embedding, but only if distinct states are separated by observable events.

Definition 5.4 (Discriminability)**.** The query system *discriminates points* if for every $\omega \neq \omega'$ in Ω there exists $E \in \mathcal{C}$ with $\omega \in E$ and $\omega' \notin E$.

Discriminability ensures the Stone embedding is injective: without it, distinct states would be identified in the completion, and the pullback of the measure on $\mathrm{St}(\mathcal{C})$ would not recover a well-defined measure on Ω.

Let $\mathrm{pure} : \Omega \to \mathrm{St}(\mathcal{C})$ denote the Stone embedding, which sends $\omega \in \Omega$ to its principal ultrafilter $\mathrm{pure}(\omega) = \{E \in \mathcal{C} : \omega \in E\}$. For $E \in \mathcal{C}$, write $[E] = \{u \in \mathrm{St}(\mathcal{C}) : E \in u\}$ for the corresponding basic clopen in $\mathrm{St}(\mathcal{C})$.

Proposition 5.5 (Structural identification)**.** *Suppose the query system discriminates points. Then*

$$\mathrm{pure}^{-1}\big(\mathcal{B}_{\mathrm{Ba}}(\mathrm{St}(\mathcal{C}))\big) \;=\; \sigma(\mathcal{C}),$$

where $\mathcal{B}_{\mathrm{Ba}}$ denotes the Baire σ-algebra.

Proof. **($\supseteq$):** For any $E = \mathrm{Cyl}(i, A) \in \mathcal{C}$, $\mathrm{pure}^{-1}([E]) = \{\omega : E \in \mathrm{pure}(\omega)\} = \{\omega : \omega \in E\} = E$. Since $[E]$ is clopen, hence Baire, every cylinder set is in $\mathrm{pure}^{-1}(\mathcal{B}_{\mathrm{Ba}})$, and therefore $\sigma(\mathcal{C}) \subseteq \mathrm{pure}^{-1}(\mathcal{B}_{\mathrm{Ba}})$.

($\subseteq$): The Stone space $\mathrm{St}(\mathcal{C})$ is compact and totally disconnected, so its clopens form a topological basis. The clopens are exactly the sets $[E]$ for $E \in \mathcal{C}$ [13, I.3.2], so the Baire σ-algebra of $\mathrm{St}(\mathcal{C})$ is $\sigma(\{[E] : E \in \mathcal{C}\})$. For any Baire set V, $\mathrm{pure}^{-1}(V) \in \sigma(\{\mathrm{pure}^{-1}([E]) : E \in \mathcal{C}\}) = \sigma(\mathcal{C})$.

Injectivity. Discriminability (Definition 5.4) implies pure is injective: if $\omega \neq \omega'$, there exists $E \in \mathcal{C}$ with $\omega \in E$ and $\omega' \notin E$, so $E \in \mathrm{pure}(\omega)$ but $E \notin \mathrm{pure}(\omega')$. Injectivity ensures the pullback does not conflate distinct events. □

Remark 5.6. On a compact Hausdorff space the Baire σ-algebra is contained in the Borel σ-algebra, with equality when the space is second-countable. We work throughout with the Baire σ-algebra on $\mathrm{St}(\mathcal{C})$; this suffices because the measure $\hat{\mu}$ constructed above is a Baire measure (it is the unique extension of a charge on the clopen algebra), and Choksi's theorem applies to Baire measures on compact spaces.

5.4 CE as support condition

Proposition 5.7 (Support condition)**.** *Let μ be a charge on $\mathcal{C}$ satisfying CE. Then the corresponding Baire measure $\hat{\mu}$ on $\mathrm{St}(\mathcal{C})$ is supported on $\mathrm{pure}(\Omega)$, the set of principal ultrafilters.*

Proof. By the Yosida–Hewitt theorem [24, 20], every bounded charge on a Boolean algebra decomposes uniquely as $\mu = \mu_c + \mu_p$, where μ_c is countably additive and μ_p is purely finitely additive. Under Stone duality, μ_c corresponds to mass on the principal ultrafilters $\mathrm{pure}(\Omega) \subset \mathrm{St}(\mathcal{C})$, while μ_p corresponds to mass on the non-principal ultrafilters [20], Ch. 10.

CE is equivalent to $\mu_p = 0$: CE requires $\mu(E_n) \to 0$ whenever $E_n \searrow \emptyset$, which is exactly the condition that the purely finitely additive part vanishes (Theorem 4.10). Therefore $\hat{\mu}_p = 0$ and $\hat{\mu}$ is supported on $\mathrm{pure}(\Omega)$. □

5.5 The Stone observational extension

Theorem 5.8 (Stone observational extension)**.** *Let the query system satisfy Surjective Evaluation, Discriminability, and common coarsenings, and let $\{\mu_i\}$ be a compatible family of normalised charges on $\{B_i\}$ satisfying CE. Then there exists a unique probability measure P on $(\Omega, \sigma(\mathcal{C}))$ such that*

$$P(\mathrm{Cyl}(i, A)) \;=\; \mu_i(\mathrm{Cyl}(i, A)) \qquad \textit{for all } i \in \iota \textit{ and } A \in \mathcal{O}_i.$$

Proof. The proof assembles the preceding propositions:

$$\{\mu_i\} \xrightarrow{\S 5.2,\,\mathrm{Step}\ 1} \{\hat{\mu}_i\} \xrightarrow{\S 5.2,\,\mathrm{Step}\ 3} \hat{\mu} \text{ on } \varprojlim_i \mathrm{St}(B_i) \xrightarrow{\text{Prop. 5.5}} P \text{ on } \sigma(\mathcal{C}).$$

Existence. By Proposition 5.2, $\mathrm{St}(\mathcal{C}) \cong \varprojlim_i \mathrm{St}(B_i)$. By Proposition 5.3, the compatible family $\{\mu_i\}$ extends to a Baire probability measure $\hat{\mu}$ on $\varprojlim_i \mathrm{St}(B_i) \cong \mathrm{St}(\mathcal{C})$.

By Proposition 5.7, $\hat{\mu}$ is supported on $\mathrm{pure}(\Omega)$. Since pure is injective (discriminability), the pushforward $P = (\mathrm{pure})_*\hat{\mu}$ — the image measure under pure^{-1} on the support — is a well-defined probability measure on Ω.

By Proposition 5.5, pure^{-1} maps Baire sets in $\mathrm{St}(\mathcal{C})$ to sets in $\sigma(\mathcal{C})$, so P is defined on $\sigma(\mathcal{C})$. For any cylinder set $E = \mathrm{Cyl}(i, A)$,

$$P(E) \;=\; \hat{\mu}([E]) \;=\; \mu_i(\mathrm{Cyl}(i, A)).$$

Uniqueness. Two probability measures on $(\Omega, \sigma(\mathcal{C}))$ agreeing on all cylinder sets are equal by Proposition 3.1, which requires common coarsenings. The Stone theorem therefore assumes common coarsenings in addition to surjective evaluation, discriminability, and CE. □

It may appear that compactness of the Stone space eliminates the need for CE. Compactness yields a σ-additive Baire measure $\hat{\mu}$ on $\mathrm{St}(\mathcal{C})$, not yet on Ω. Descent to Ω is equivalent to $\hat{\mu}$ being supported on $\mathrm{pure}(\Omega)$, the principal ultrafilters; and by Proposition 5.7 this support condition is exactly CE. In the Stone picture, CE is not an additional hypothesis but a *support condition*: it is what ensures the measure concentrates on realised states rather than ideal limit points.

6 Synthesis

The results of this paper separate three layers: coherence, completion, and admissibility. A finitely additive charge gives a coherent valuation on the Boolean algebra of observable distinctions; Stone duality completes that algebra into the space of all coherent distinction patterns, including nonprincipal ones; and collective exhaustion is the condition that determines when such a valuation extends to genuine probability. In this sense, the gap between finite coherence and probability is the gap between unrestricted support on the Stone completion and support that remains honest under observational exhaustion.

The Yosida–Hewitt decomposition makes this precise: $\mu_p = 0$ is both the vanishing of the purely finitely additive part and the condition that no mass escapes to non-principal ultrafilters.

When the index set admits sequential common refinements and the query system discriminates points, both constructions are available simultaneously. The following proposition confirms they produce the same observable measure.

Proposition 6.1 (Uniqueness of the observable measure)**.** *Suppose the query system satisfies Surjective Evaluation, Discriminability, and sequential common refinements and common coarsenings, and let $\{\nu_i\}$ be a compatible family of probability measures. Let P^{C} be the measure produced by Theorem 3.5 and P^{S} the measure produced by Theorem 5.8. Then $P^{\mathrm{C}} = P^{\mathrm{S}}$.*

Proof. Both measures agree on every cylinder set: $P^{\mathrm{C}}(\mathrm{Cyl}(i, A)) = \nu_i(A) = P^{\mathrm{S}}(\mathrm{Cyl}(i, A))$. By observational determination (Proposition 3.1), they are equal on $\sigma(\mathcal{C})$. $\square$

Corollary 6.2. *The observable measure may be obtained by two distinct constructions: by countable reach in ι via sequential common refinements (Theorem 3.5), or by the Stone completion of $\mathcal{C}$ (Theorem 5.8). Whenever both constructions apply, they yield the same measure.*

The Stone space makes the geometry of that demand visible. Every point of $\mathrm{St}(\mathcal{C})$ is a maximal consistent assignment of yes/no answers to every observable distinction — a coherent distinction pattern, whether or not any $\omega \in \Omega$ realises it. The realised instantiation sits inside $\mathrm{St}(\mathcal{C})$ as the principal ultrafilters; beyond it lie ideal limit points, internally coherent but unrealised.

The measure now exists. The next question is what it implies: whether prediction, dynamics, and ultimately the recovery of the underlying state space follow from observation alone — or whether they too must be assumed.

Part II
Dynamics and Reconstruction from Observation

Abstract

Starting from a σ-additive probability measure on an observable algebra [19], this paper derives two structures that are often assumed: the dynamics of prediction, and the recoverability of the state space.

Making prediction precise forces the introduction of the predictive kernel $\Pi_Q : \alpha \to \mathcal{P}(\beta)$, the regular conditional distribution of a future observation F given a current observation Q. Composing with Q gives the minimal predictive state map $Q_* : \Omega \to \mathcal{P}(\beta)$. Indexing over time, the kernels compose via Chapman–Kolmogorov — derived, not assumed — yielding a Markov semigroup $\{K_t\}$ and the Koopman–Perron duality $\int K_t g \, d\mu = \int g \, d(\mu \star \Pi_t)$. When the kernels are Dirac measures, K_t collapses to the classical Koopman operator.

The construction produces a predictive state space $\mathcal{P}(\beta)$. In a measure-preserving system $(X, \mathcal{B}, \mu, T)$ with $Q_t = h \circ T^t$, the map Q_* becomes the delay map $\Phi_h(x) = (h(T^n x))_{n \in \mathbb{Z}}$. The reconstruction question is then internal to the same framework: whether this predictive state is faithful — whether the prediction machinery sees all of X, or only a proper quotient.

The answer is a three-way equivalence — the reconstruction theorem. State-space recovery ($\mathcal{O}_h = \mathcal{B} \bmod \mu$) holds if and only if the delay algebra $\mathcal{A}_h$ is L^2-dense, if and only if Φ_h is a measure-theoretic embedding. The bridge between the first two conditions is a general lemma: for any subalgebra $\mathcal{A} \subseteq L^\infty$ containing 1, the L^2-closure of $\mathcal{A}$ equals $L^2(X, \sigma(\mathcal{A}), \mu)$. When reconstruction holds, the compact Stone space [19] — built as a completion of the sample space for deriving σ-additivity — is identified with X itself.

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1 Introduction

The classical approach to dynamics, following Koopman [15], describes how observable functions evolve under a known transformation T: the operator $U_T f = f \circ T$ acts by precomposing with T as a primitive. This paper asks two questions that precede this picture.

The first is: given a current observation $Q : \Omega \to \alpha$ and a future observation $F : \Omega \to \beta$, what does an observer predict about F given Q? The dynamics, if there are any, should follow from the answer — not precede it. Making prediction precise immediately requires a σ-additive probability measure on the observable algebra. That measure is not an extra assumption: a coherent family of finitely additive charges on a directed system of observable algebras extends to a genuine probability measure exactly when it satisfies collective exhaustion [19], the condition ruling out persistent mass on globally vanishing events.

Once the measure exists, the canonical answer to the prediction question is the regular conditional distribution of F given Q: the unique Markov kernel $\Pi_Q : \alpha \to \mathcal{P}(\beta)$ realising $P(F \in \cdot \mid Q = q)$ simultaneously across all outcomes. Its existence rests on the Rokhlin disintegration theorem [21], which requires σ-additivity. Composing with Q gives the minimal predictive state map $Q_* : \Omega \to \mathcal{P}(\beta)$, the minimal sufficient statistic for future prediction. Indexing over time, the kernels $\{\Pi_t\}$ compose; their composition rule is the Chapman–Kolmogorov equation, derived from temporal coherence rather than assumed. From it emerge a semigroup of Markov operators

$\{K_t\}$ and the Koopman–Perron duality connecting them with the pushforward operators. When the kernels are Dirac measures, the construction collapses to the classical Koopman picture.

The construction produces a predictive state space $\mathcal{P}(\beta)$ and a stationary distribution ν_{Q_*}. In a measure-preserving system $(X, \mathcal{B}, \mu, T)$ with $Q_t = h \circ T^t$ for some $h \in L^\infty$, the map Q_* specialises to the delay map $\Phi_h(x) = (h(T^n x))_{n \in \mathbb{Z}}$. The reconstruction question is therefore not a separate problem: it asks whether this predictive state is faithful — whether the prediction machinery sees all of X, or only a proper quotient. The answer is a three-way equivalence: the σ-algebraic condition $\mathcal{O}_h = \mathcal{B}$ modulo μ, the analytic condition that the delay algebra $\mathcal{A}_h$ is L^2-dense, and the geometric condition that Φ_h is a measure-theoretic embedding are all equivalent.

The key to the equivalence is the density bridge lemma: for any subalgebra $\mathcal{A} \subseteq L^\infty(X, \mu)$ containing the constant 1, the L^2-closure of $\mathcal{A}$ equals $L^2(X, \sigma(\mathcal{A}), \mu)$. The proof goes via the functional monotone class theorem [6]. This is what lets the σ-algebraic condition ($\mathcal{O}_h = \mathcal{B}$ mod μ) and the analytic condition ($\mathcal{A}_h$ L^2-dense) pass freely between each other — they are not two separate conditions but two descriptions of the same constraint.

When reconstruction holds, the Stone space [19] $\mathrm{St}(\mathcal{O}_h)$ — built as a compact completion of the sample space for deriving σ-additivity — turns out to be measure-theoretically isomorphic to X itself. The programme that began by seeking a compact ambient space for the measure ends by identifying it with the state space being observed.

The paper is self-contained modulo the probability measure. Sections 2 and 3 develop the predictive structure from a probability space $(\Omega, \mathcal{F}, P)$. Section 4 makes the connection explicit: specialising Q_* to a measure-preserving system yields the delay map, and the reconstruction question becomes whether that map is injective modulo null sets. Sections 5 and 6 carry out the reconstruction, proving the density bridge and the reconstruction theorem.

2 Prediction and the minimal predictive state

2.1 Predictive kernel

Fix a probability space $(\Omega, \mathcal{F}, P)$ and two measurable maps:

- $Q : \Omega \to \alpha$, where $(\alpha, \mathcal{A})$ is a measurable space;
- $F : \Omega \to \beta$, where $(\beta, \mathcal{B})$ is a standard Borel space.

The standard Borel hypothesis on β is used once: it is the hypothesis under which the Rokhlin disintegration theorem [21] guarantees the existence of a regular conditional distribution. All other results hold in greater generality.

We write $\mathcal{P}(\beta)$ for the space of probability measures on β, equipped with its canonical (Giry) σ-algebra. The joint pushforward $\rho := P \circ (Q, F)^{-1}$ on $\alpha \times \beta$ has marginals $\rho_\alpha = P \circ Q^{-1}$ and $\rho_\beta = P \circ F^{-1}$.

The regular conditional distribution of F given Q is the unique (up to $P \circ Q^{-1}$-null sets) Markov kernel $\Pi_Q : \alpha \to \mathcal{P}(\beta)$ satisfying

$$\int_A \Pi_Q(q, B)\, d(P \circ Q^{-1})(q) \;=\; P(Q \in A,\ F \in B)$$

for all measurable $A \subseteq \alpha$ and $B \subseteq \beta$.

Definition 2.1 (Predictive kernel)**.** The *predictive kernel* is Π_Q, the disintegration kernel of $\rho = P \circ (Q, F)^{-1}$ over its first marginal: $\Pi_Q(q, A) = P(F \in A \mid Q = q)$.

The kernel is covariant under measurable maps: if $\pi : \alpha \to \alpha'$ is measurable and $Q' = \pi \circ Q$, then $\Pi_Q(q, \cdot) = \Pi_{Q'}(\pi(q), \cdot)$ for $(P \circ Q^{-1})$-almost every q, by uniqueness of disintegration.

Composing Π_Q with Q gives the map sending each sample point to its conditional law.

2.2 Minimal predictive state

Definition 2.2 (Minimal predictive state map)**.** The *minimal predictive state map* is

$$Q_* : \Omega \to \mathcal{P}(\beta), \qquad Q_*(\omega) \;=\; \Pi_Q(Q(\omega), \cdot).$$

Since Π_Q is a Markov kernel it is measurable, and Q_* inherits measurability by composition.

Theorem 2.3 (Predictive factorization)**.** *For every bounded measurable $g : \beta \to \mathbb{R}$:*

$$E[g(F) \mid \sigma(Q)] \;=\; \int g\, dQ_*(\cdot) \quad P\text{-a.e.}$$

Proof. Let $h(\omega) := \int g\, dQ_*(\omega)$. We verify the two defining properties.

Measurability. The map $q \mapsto \int g\, d\Pi_Q(q, \cdot)$ is strongly measurable; since $Q_* = \Pi_Q(Q(\cdot), \cdot)$, the map h is $\sigma(Q)$-measurable.

Set-integral identity. For $S = Q^{-1}(A)$:

$$\begin{aligned}\int_S h\, dP &= \int_A \int g\, d\Pi_Q(q, \cdot)\, d(P \circ Q^{-1})(q) \quad \text{(pushforward)}\\ &= \int_{A\times\beta} g(f)\, d\rho(q, f) \quad \text{(disintegration)}\\ &= \int_S g(F)\, dP.\end{aligned}$$

The conclusion follows from a.e. uniqueness of conditional expectation. □

Corollary 2.4 (Predictive sufficiency)**.** *$E[g(F) \mid \sigma(Q)] = E[g(F) \mid \sigma(Q_*)]$ P-a.e.*

Proof. $\sigma(Q_*) \subseteq \sigma(Q)$, and the conditional expectation given $\sigma(Q)$ is already $\sigma(Q_*)$-measurable by the theorem. □

The operator $(K_{Q_*} g)(q_*) = \int g\, dq_*$ on $\mathcal{P}(\beta)$ recasts the theorem as: $E[g(F) \mid \sigma(Q)] = (K_{Q_*} g) \circ Q_*$ P-a.e.

3 Dynamics and duality

3.1 Markov semigroup

To introduce dynamics, index over a discrete time horizon. For each $t \in \mathbb{N}$, let $\Pi_t : \gamma \to \mathcal{P}(\gamma)$ be a Markov kernel on the state space γ, with $\Pi_0(q, \cdot) = \delta_q$. A family $\{\Pi_t\}$ satisfying the Chapman–Kolmogorov equation

$$\Pi_{t+s}(q, \cdot) \;=\; \int_\gamma \Pi_t(q', \cdot)\, d\Pi_s(q, q') \quad \text{(i.e., } \Pi_{t+s} = \Pi_t \circ_\kappa \Pi_s\text{)}$$

is a *Markov semigroup kernel.* The associated *Markov operators* are

$$(K_t g)(q) \;:=\; \int g\, d\Pi_t(q, \cdot), \qquad g : \gamma \to \mathbb{R}.$$

Each K_t is positive, unital, and contractive on bounded measurable functions, with $K_0 = \mathrm{id}$; positivity, unitality, and contractiveness are immediate from the Markov property.

Proposition 3.1 (Semigroup property)**.** *$K_{t+s} g = K_t(K_s g)$ for all bounded measurable g.*

Proof. $(K_{t+s} g)(q) = \int g\, d(\Pi_t \circ_\kappa \Pi_s)(q, \cdot) = \int \left(\int g\, d\Pi_t(q', \cdot) \right) d\Pi_s(q, q') = \int (K_t g)(q')\, d\Pi_s(q, q') = (K_t(K_s g))(q)$, where the exchange of integrals is justified by Bochner–Fubini. □

The pushforward of a measure μ on γ through the kernel is $P_t^* \mu := \mu \star \Pi_t = \int \Pi_t(q, \cdot)\, d\mu(q)$.

3.2 Koopman–Perron duality

Theorem 3.2 (Koopman–Perron duality)**.** *For every bounded measurable g and finite measure μ:*

$$\int (K_t g)\, d\mu \;=\; \int g\, d(P_t^* \mu).$$

Proof. $\int g\, d(P_t^*\mu) = \int g\, d(\mu \star \Pi_t) = \int_\gamma \int g\, d\Pi_t(q,\cdot)\, d\mu(q) = \int (K_t g)\, d\mu$, by Bochner–Fubini. $\square$

When $\Pi_t(q,\cdot) = \delta_{\varphi_t(q)}$ for a measurable map $\varphi_t : \gamma \to \gamma$, the stochastic picture collapses.

3.3 Specialisation to measure-preserving systems

Proposition 3.3 (Deterministic limit)**.** *If $\Pi_t(q,\cdot) = \delta_{\varphi_t(q)}$, then $K_t g = g \circ \varphi_t$ and $\varphi_{t+s} = \varphi_t \circ \varphi_s$.*

Proof. $(K_t g)(q) = \int g\, d\delta_{\varphi_t(q)} = g(\varphi_t(q))$. The flow law follows from $\delta_{\varphi_{t+s}(q)} = (\Pi_t \circ_\kappa \Pi_s)(q,\cdot) = \delta_{\varphi_t(\varphi_s(q))}$ and injectivity of the Dirac embedding. $\square$

In the deterministic case, $K_t g = g \circ \varphi_t = U_T^t g$: the Markov operator specialises to the classical Koopman operator. The Koopman–Perron duality $\int K_t g\, d\mu = \int g\, d(P_t^*\mu)$ becomes the change-of-variables formula $\int g \circ T^t\, d\mu = \int g\, d(T_*^t \mu)$.

Example 3.4 (Finite Markov chain)**.** Let $X = \{1, \dots, k\}$, $\mathcal{B} = 2^X$, and let $P = (p_{ij})$ be the transition matrix of a finite irreducible Markov chain with unique stationary distribution μ ($\mu P = \mu$). Take full-state observation $Q_t = X_t$. The predictive kernel is $\Pi(i,\cdot) = P(i,\cdot)$, the i-th row of P: this is the unique Markov kernel satisfying $\mathbb{E}_\mu[f(X_1) \mid X_0 = i] = \sum_j p_{ij} f(j)$ for all bounded f. The t-step kernel is $\Pi^{(t)}(i,\cdot) = P^t(i,\cdot)$, and temporal coherence gives $\Pi^{(t+s)} = \Pi^{(t)} \circ_\kappa \Pi^{(s)}$, i.e. $P^{t+s} = P^t P^s$. The semigroup acts as

$$K_t f(i) = \sum_j p_{ij}^t f(j).$$

Koopman–Perron duality reduces to the stationary-measure identity:

$$\int K_t g\, d\mu = \mu^\top P^t g = (\mu P^t)^\top g = \mu^\top g = \int g\, d\mu,$$

the fixed-point property $\mu P = \mu$ of the stationary distribution. In this finite setting every charge is automatically σ-additive, so the admissibility condition of [19] is trivially satisfied. Reconstruction holds at lag 0: $\mathcal{O}_h = \sigma(h) = \mathcal{B}(X)$ whenever h separates points of X.

The full structure produced by the prediction construction is captured in one definition.

Definition 3.5 (Observable dynamical system)**.** An *observable dynamical system* is a triple $(\mathcal{P}(\beta), \nu_{Q_*}, \{K_t\}_{t\in\mathbb{N}})$ where $\mathcal{P}(\beta)$ is the predictive state space (the image of Q_*), $\nu_{Q_*} = P \circ Q_*^{-1}$ is the pushforward of P along Q_*, and $\{K_t\}$ is the Markov semigroup on $\mathcal{P}(\beta)$.

4 From prediction to reconstruction

Setting $Q_t = h \circ T^t$ in a measure-preserving system $(X, \mathcal{B}, \mu, T)$ with $h \in L^\infty$ specialises the construction. Under this choice, $Q_*(\omega) = (h(T^n\omega))_{n\in\mathbb{Z}}$ is the delay map $\Phi_h(x) = (h(T^n x))_{n\in\mathbb{Z}}$, and the observable dynamical system is the triple $(\Phi_h(X), \mu \circ \Phi_h^{-1}, \{K_t\})$.

The predictive state space $\Phi_h(X)$ is a subset of $\mathbb{R}^\mathbb{Z}$ encoding the dynamical information that h can see. Whether it is a faithful image of X depends on how much the delay orbit

$\{h \circ T^n : n \in \mathbb{Z}\}$ collectively distinguishes points of X. Injectivity of Φ_h modulo μ-null sets is equivalent to the observable σ-algebra

$$\mathcal{O}_h = \sigma\big(\{h \circ T^n : n \in \mathbb{Z}\}\big)$$

equalling $\mathcal{B}(X)$ modulo μ.

The Koopman–Perron duality (Theorem 3.2) connects this to a function-space question: the observable σ-algebra $\mathcal{O}_h$ determines the Markov semigroup through the algebra it generates, and the reconstruction question is precisely whether that algebra is large enough to recover all of $\mathcal{B}$. The next two sections answer this question.

5 Density bridge and reconstruction

5.1 Density bridge lemma

Throughout this section and the next, $(X, \mathcal{B}, \mu)$ is a standard probability space, $T : X \to X$ is a measurable, μ-preserving invertible bimeasurable bijection with $T_*\mu = \mu$, and $h \in L^\infty(X, \mu)$.

Definition 5.1 (Observable algebra)**.** The *observable algebra* generated by h is

$$\mathcal{O}_h \;:=\; \sigma\big(\{h \circ T^n : n \in \mathbb{Z}\}\big) \;\subseteq\; \mathcal{B}(X).$$

The *delay algebra* $\mathcal{A}_h$ is the smallest subalgebra of $L^\infty(X, \mu)$ containing 1 and every $h \circ T^n$; concretely, finite sums $\sum_\alpha c_\alpha \prod_k (h \circ T^{n_k})^{e_k}$.

Definition 5.2 (Delay map)**.** The *delay map* is

$$\Phi_h : X \to \mathbb{R}^{\mathbb{Z}}, \qquad \Phi_h(x) \;=\; \big(h(T^n x)\big)_{n\in\mathbb{Z}}.$$

Since $\pi_n \circ \Phi_h = h \circ T^n$ for each coordinate projection π_n, the map Φ_h is measurable and $\mathcal{O}_h = \Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}}))$. Thus $\mathcal{O}_h = \mathcal{B} \bmod \mu$ is precisely the condition that Φ_h is a measure-theoretic embedding.

The central lemma connects the Boolean algebra level to the L^2 level. It holds for any subalgebra of L^∞.

Lemma 5.3 (Density bridge)**.** *Let $(X, \mathcal{B}, \mu)$ be a probability space and let $\mathcal{A} \subseteq L^\infty(X, \mu)$ be a subalgebra containing the constant function* 1*. Then:*

$$\overline{\mathcal{A}}^{L^2} \;=\; L^2(X,\, \sigma(\mathcal{A}),\, \mu).$$

Consequently, $\overline{\mathcal{A}}^{L^2} = L^2(X, \mu)$ if and only if $\sigma(\mathcal{A}) = \mathcal{B}$ modulo μ.

Proof. Write $\mathcal{M} := \sigma(\mathcal{A})$ and $V := \overline{\mathcal{A}}^{L^2}$.

Step 1: $V \subseteq L^2(X, \mathcal{M}, \mu)$**.** Every $f \in \mathcal{A}$ is $\mathcal{A}$-measurable, hence $\mathcal{M}$-measurable. So $\mathcal{A} \subseteq L^2(X, \mathcal{M}, \mu)$. Since $L^2(X, \mathcal{M}, \mu)$ is a closed subspace of $L^2(X, \mathcal{B}, \mu)$, taking the closure gives $V \subseteq L^2(X, \mathcal{M}, \mu)$.

Step 2: $L^2(X, \mathcal{M}, \mu) \subseteq V$**.** It suffices to show that every indicator $\mathbf{1}_E$ with $E \in \mathcal{M}$ lies in V. By the functional monotone class theorem [6], the class

$$\mathcal{H} \;:=\; \big\{f \in L^\infty(X, \mathcal{M}, \mu) : f \in V\big\}$$

is a vector space containing $\mathcal{A}$ and closed under bounded monotone pointwise limits (since if $f_n \in V$ are uniformly bounded and $f_n \to f$ pointwise a.e. with $f \in L^\infty$, then $f_n \to f$ in L^2 by dominated convergence, so $f \in V$). Since $\mathcal{A}$ is an algebra containing 1 and $\sigma(\mathcal{A}) = \mathcal{M}$, the monotone class theorem gives $\mathcal{H} = L^\infty(X, \mathcal{M}, \mu)$. In particular every $\mathcal{M}$-indicator $\mathbf{1}_E \in V$.

Since the $\mathcal{M}$-simple functions are dense in $L^2(X, \mathcal{M}, \mu)$ and V is closed, $L^2(X, \mathcal{M}, \mu) \subseteq V$.

Conclusion. Steps 1 and 2 give $V = L^2(X, \mathcal{M}, \mu)$. The final equivalence is immediate: $V = L^2(X, \mathcal{B}, \mu)$ iff $L^2(X, \mathcal{M}, \mu) = L^2(X, \mathcal{B}, \mu)$ iff $\mathcal{M} = \mathcal{B} \bmod \mu$. ☐

Remark 5.4 (Role of the algebra structure)**.** The algebra structure of $\mathcal{A}$ is essential. The corresponding statement for the closed *linear* span $\mathcal{H}_{\mathcal{A}} = \overline{\mathrm{span}}(\mathcal{A})$ is false in general: $\mathcal{H}_{\mathcal{A}} = L^2$ does not imply $\sigma(\mathcal{A}) = \mathcal{B}$.

For example, take $X = [0, 1]$, μ = Lebesgue, $T(x) = 2x \bmod 1$ (the doubling map), and $h(x) = x$. The orbit $\{h \circ T^n\}(x) = \{2^n x \bmod 1\}$ generates $\mathcal{B}([0, 1])$ as a σ-algebra (the dyadic structure separates points). But $\mathrm{span}\{2^n x \bmod 1 : n \geq 0\}$ is a proper subspace of $L^2[0, 1]$: it does not contain x^2. So the linear span can be a proper dense subspace even when $\sigma(\mathcal{A}) = \mathcal{B}$.

In the other direction, h cyclic for U_T (linear span dense) always implies $\sigma(\mathcal{A}_h) = \mathcal{B} \bmod \mu$ — see Corollary 6.2.

5.2 Reconstruction theorem

Theorem 5.5 (Reconstruction theorem)**.** *Let $(X, \mathcal{B}, \mu, T)$ be an invertible measure-preserving system and $h \in L^\infty(X, \mu)$. The following are equivalent:*

(i) $\mathcal{O}_h = \mathcal{B}$ modulo μ;

(ii) $\mathcal{A}_h$ is dense in $L^2(X, \mu)$;

(iii) $\Phi_h : X \to \mathbb{R}^{\mathbb{Z}}$ is a measure-theoretic embedding: $\Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}})) = \mathcal{B}$ modulo μ.

Proof. (i) $\Leftrightarrow$ (ii). Apply the density bridge (Lemma 5.3) with $\mathcal{A} = \mathcal{A}_h$ and $\sigma(\mathcal{A}_h) = \mathcal{O}_h$: $\overline{\mathcal{A}_h}^{L^2} = L^2(X, \mu)$ iff $\mathcal{O}_h = \mathcal{B} \bmod \mu$.

(i) $\Leftrightarrow$ (iii). By Definition 5.2, $\mathcal{O}_h = \Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}}))$. So $\mathcal{O}_h = \mathcal{B} \bmod \mu$ iff $\Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}})) = \mathcal{B} \bmod \mu$, which is exactly condition (iii). □

Remark 5.6 (What "measure-theoretic embedding" means)**.** Condition (iii) says that Φ_h is *almost everywhere injective* and the pullback σ-algebra is all of $\mathcal{B}$. More precisely: since $\Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}})) = \mathcal{B} \bmod \mu$, for any two sets $A, A' \in \mathcal{B}$ with $\mu(A \triangle A') > 0$ there exists a Borel set $E \subseteq \mathbb{R}^{\mathbb{Z}}$ with $\mu(\Phi_h^{-1}(E) \triangle A) = 0$ and $\mu(\Phi_h^{-1}(E) \triangle A') > 0$. In measure-theoretic language, Φ_h induces an isomorphism of measure algebras $(X/\mu, \mathcal{B}/\mu) \cong (\mathbb{R}^{\mathbb{Z}}/(\Phi_h)_*\mu, \mathcal{B}(\mathbb{R}^{\mathbb{Z}})/(\Phi_h)_*\mu)$.

The T-invariance of μ implies that $(\Phi_h)_*\mu$ is invariant under the bilateral shift σ on $\mathbb{R}^{\mathbb{Z}}$, since $(\Phi_h(Tx))_n = h(T^{n+1}x) = (\sigma(\Phi_h(x)))_n$ for each n. The dynamical system (X, T, μ) is therefore isomorphic to $(\mathbb{R}^{\mathbb{Z}}, \sigma, (\Phi_h)_*\mu)$ restricted to the image of Φ_h.

6 Cyclic vectors and Stone space identification

6.1 Cyclic vectors

Definition 6.1 (Cyclic vector)**.** The function $h \in L^2(X, \mu)$ is a *cyclic vector* for U_T if

$$\mathcal{H}_h \;:=\; \overline{\mathrm{span}}\big\{U_T^n h : n \in \mathbb{Z}\big\} \;=\; L^2(X, \mu).$$

Corollary 6.2 (Cyclic implies reconstruction)**.** *If $h \in L^\infty(X, \mu)$ is a cyclic vector for U_T, then $\mathcal{O}_h = \mathcal{B}$ modulo μ.*

Proof. Since $\mathcal{A}_h$ contains $\mathrm{span}\{h \circ T^n\} = \mathrm{span}\{U_T^n h\}$ as a subset, $\overline{\mathcal{A}_h}^{L^2} \supseteq \mathcal{H}_h = L^2$. By the density bridge, $\mathcal{O}_h = \mathcal{B} \bmod \mu$. □

Remark 6.3 (The implication is strict)**.** The converse of Corollary 6.2 fails in general: $\mathcal{O}_h = \mathcal{B}$ does not imply h is cyclic. The doubling map example of Remark 5.4 witnesses this.

The cyclic vector condition asks for density of the *linear span* of the orbit; reconstruction asks only for density of the *algebra*. For a bounded h, the linear span is contained in the algebra, so cyclicity is the stronger condition.

The two conditions coincide precisely when U_T has *simple L^2-spectrum*: $L^2(X,\mu)$ is a cyclic U_T-module, meaning there exists some cyclic vector. In this case, every generating observation h (satisfying $\mathcal{O}_h = \mathcal{B}$) is automatically cyclic. Simple spectrum is a generic property of ergodic transformations in the Baire category sense [10] but is not universal.

6.2 Stone space identification

The Stone space $\mathrm{St}(\mathcal{O}_h)$ of the observable algebra is the compact Hausdorff space whose points are ultrafilters on $\mathcal{O}_h$, with the natural Stone embedding $\iota : X \to \mathrm{St}(\mathcal{O}_h)$ sending x to the principal ultrafilter $\{E \in \mathcal{O}_h : x \in E\}$.

Proposition 6.4 (Stone space identification)**.** *Suppose $\mathcal{O}_h = \mathcal{B}$ modulo μ. Then ι is a measure-theoretic isomorphism: ι identifies the measure algebra $(X/\mu, \mathcal{B}/\mu)$ with $(\mathrm{St}(\mathcal{O}_h)/\iota_*\mu, \mathcal{B}(\mathrm{St}(\mathcal{O}_h))/\iota_*\mu)$.*

Proof. Injectivity mod μ. Suppose $\iota(x) = \iota(x')$. For any $E \in \mathcal{B}$ with $x \in E \not\ni x'$, find $F \in \mathcal{O}_h$ with $\mu(E \triangle F) = 0$. Since $x \in F$ and $\iota(x) = \iota(x')$, also $x' \in F$, so $x' \in (E\triangle F) \cup F^c$, placing x' in a null set. Hence ι is injective μ-a.e.

σ-algebra identification. The pullback ι^{-1} maps the Baire σ-algebra of $\mathrm{St}(\mathcal{O}_h)$ (generated by the clopen sets $[E] = \{u : E \in u\}$) to $\mathcal{O}_h = \mathcal{B}$ mod μ.

Measure recovery. Since μ is σ-additive on $\mathcal{O}_h = \mathcal{B}$, the collective exhaustion condition of [19] forces $\iota_*\mu$ to concentrate on the principal ultrafilters $\iota(X)$, so $\iota_*\mu$ is supported on the image and the measure algebras are identified. □

The Stone space [19] — built as a compact ambient space for deriving σ-additivity — is, under the reconstruction condition, the state space X itself. The construction that began by seeking a compact completion of the sample space ends by recognising it as the object the observer was trying to recover.

Example 6.5 (Cantor set: reconstruction without smoothness)**.** Let $\mathcal{C} \subset [0,1]$ be the middle-thirds Cantor set, and let $\mu_{\mathcal{C}}$ be the Cantor measure — the pushforward of the fair-coin product measure $\nu = (\frac{1}{2}\delta_0 + \frac{1}{2}\delta_2)^{\mathbb{N}}$ under the ternary coding map $\pi(d_1, d_2, \ldots) = \sum_{k\geq 1} d_k/3^k$. Let $S : \mathcal{C} \to \mathcal{C}$ be the shift map

$$S(x) = \begin{cases} 3x & x \in \mathcal{C} \cap [0, \frac{1}{3}], \\ 3x - 2 & x \in \mathcal{C} \cap [\frac{2}{3}, 1], \end{cases}$$

and let $h(x) = x$. The system $(\mathcal{C}, \mu_{\mathcal{C}}, S)$ is measurably conjugate to the Bernoulli shift $(\{0,2\}^{\mathbb{N}}, \nu, \sigma)$ via π; in particular S is $\mu_{\mathcal{C}}$-invariant and ergodic.

For $x = \pi(d_1, d_2, \ldots)$ we have $S^n x = \pi(d_{n+1}, d_{n+2}, \ldots)$. Since every point of $\mathcal{C}$ lies in $[0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$,

$$d_{n+1} = 0 \iff S^n x \in \left[0, \tfrac{1}{3}\right], \qquad d_{n+1} = 2 \iff S^n x \in \left[\tfrac{2}{3}, 1\right],$$

so $h \circ S^n$ recovers the $(n+1)$-st ternary digit of x. The delay vector $\Phi_h^{(L)}(x) = (x, Sx, \ldots, S^L x)$ reads off digits $d_1, \ldots, d_{L+1}$; its fibres are the generation-$(L+1)$ cylinder sets

$$F_z = \{x \in \mathcal{C} : d_k(x) = z_k,\ 1 \leq k \leq L+1\}, \quad z \in \{0,2\}^{L+1},$$

each carrying $\mu_{\mathcal{C}}(F_z) = 2^{-(L+1)}$, so $\mathcal{O}_h^{(L)} = \sigma(\Phi_h^{(L)}) = \sigma\{F_z : z \in \{0,2\}^{L+1}\}$. Since $\mu_{\mathcal{C}}$-a.e. pair of distinct points differs at some finite digit, these algebras separate points as $L \to \infty$:

$$\mathcal{O}_h \ = \ \sigma\Big(\bigcup_{L\geq 0} \mathcal{O}_h^{(L)}\Big) \ = \ \mathcal{B}(\mathcal{C}) \quad \text{mod } \mu_{\mathcal{C}}.$$

By Theorem 5.5, Φ_h is a measure-theoretic embedding: the finite-lag observable algebras refine to the full Borel structure as $L \to \infty$.

$\mathcal{C}$ is not a smooth manifold: it is compact and zero-dimensional, with Hausdorff dimension $\log 2/\log 3 \approx 0.63$. The differential-geometric hypotheses of Takens's theorem — C^2 ambient dynamics, integer manifold dimension d, embedding bound $2d+1$ — are unavailable before one even asks about regularity of S. Takens has no domain of application here. Theorem 5.5 applies because it requires only that the observable algebra be measurably generated; smoothness plays no role.

The classical Takens embedding theorem [23] reaches a related conclusion by a different route: for a C^2 diffeomorphism on a compact d-manifold and a generic smooth h, the finite-delay map $\Phi_h^{(k)}(x) = (h(x), h(Tx), \ldots, h(T^{k-1}x))$ is a diffeomorphic embedding into $\mathbb{R}^{2d+1}$ for $k \geq 2d+1$. The reconstruction theorem here requires only measurability, uses all delays simultaneously, and replaces the dimension count with the algebraic density condition; in exchange, the embedding is measure-theoretic rather than topological. On a compact manifold with generic smooth h, the orbit $\{h \circ T^n\}$ separates points, so Stone–Weierstrass gives $\mathcal{A}_h$ uniformly dense in $C(M)$ and hence L^2-dense: the density condition holds and the two results are compatible.

Reconstruction is an exact asymptotic condition: it asks whether the delay algebra generates all of $\mathcal{B}$ in the limit of all lags, and says nothing about rates, finite approximations, or what a finite time series can certify. Those questions are the subject of what follows.

Part III
Certifying Reconstruction from Finite Data

Abstract

We study finite-sample reconstruction in measure-preserving systems. Given a time series $x_1, \ldots, x_n$ from a system $(X, \mathcal{B}, \mu, T)$ and a single observable h, we ask whether reconstruction — the condition that the delay algebra $\sigma(h, h \circ T, \ldots)$ generates $\mathcal{B}$ modulo μ — is detectable from data alone, at what rate, and with what certificates.

We prove three theorems. The *Algebra Theorem* shows that the empirical σ-algebra approximation error $\hat{\delta}(L, n)$ concentrates around the true error $\delta(L)$, and that the elbow stopping rule $\hat{L}^*$ achieves the minimax-optimal rate $n^{-s/(2s+d)}$ for Hölder(s) targets on a d-dimensional state space, without oracle knowledge of the mixing rate, the lag, or the smoothness index. The *Dynamics Theorem* shows that under uniform separation, the delay map $\Phi_h^{(L)}$ is bi-Lipschitz and estimated delay vectors certify point separation at rate $n^{-\beta/(2\beta+d)}$. The *Conjunction Theorem* shows that for deterministic T, algebra separation and metric separation are the same event — not merely correlated, but identical — and that both witnesses certify this from data under reconstruction and uniform separation.

A third computable witness emerges from the fibre structure of the delay map. Under the fibre mixing condition — that ν_L-almost every large fibre is balanced — the collision entropy $H_2(\nu_L) := -\log(\mu \otimes \mu)(R_L)$ diverges if and only if $\delta(L) \to 0$. The key is a conditional variance identity expressing $\delta(L)$ as an integral of within-fibre variance over unseparated trajectory pairs; the identity reverses under fibre mixing but fails when large fibres are monochromatic — entirely inside or outside the hardest-to-approximate event. All three witnesses — algebraic, geometric, and information-theoretic — measure the same failure of separation across delay fibres.

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1 Introduction

The reconstruction theorem states that the observable algebra $\mathcal{O}_h = \sigma\{h \circ T^n : n \in \mathbb{Z}\}$ generates $\mathcal{B}$ modulo μ if and only if the delay map Φ_h is a measure-theoretic embedding — an exact asymptotic condition that says nothing about rates, certificates, or what can be verified from a finite time series. Given a time series $x_1, \ldots, x_n$ from $(X, \mathcal{B}, \mu, T)$ and a single observable h, we ask: can one determine from data alone whether reconstruction holds, at what rate, and with what certificates? We answer with three theorems: the Algebra Theorem, the Dynamics Theorem, and the Conjunction Theorem. Both results [19, 18] are taken as given; this paper asks what finite data can certify.

The Algebra Theorem (Section 5). Define the σ-algebra approximation error

$$\delta(L) = \sup_{S \in \mathcal{B}} \inf_{E \in \mathcal{O}_h^{(L)}} \mu(S \triangle E),$$

which measures how much of $\mathcal{B}$ is not yet captured by the finite-lag algebra $\mathcal{O}_h^{(L)} = \sigma(h, h \circ T, \ldots, h \circ T^L)$. This quantity is estimable from data as $\hat{\delta}(L, n)$ (defined in Section 2). The elbow stopping rule $\hat{L}^*$ fires when $\hat{\delta}(L, n)$ stops decreasing. Under exponential mixing with resolvability constant $C_\lambda > 4$, $\hat{L}^*$ achieves the minimax-optimal rate $n^{-s/(2s+d)}$ for Hölder(s) targets on a d-dimensional state space, without any oracle knowledge of the mixing rate, the lag $L^*(n)$, or the smoothness index s. This is the main finite-sample rate result.

The Dynamics Theorem (Section 6). The delay map $\Phi_h^{(L)}(x) = (h(x), \ldots, h(T^L x))$ encodes the trajectory of x under the observable. When $T \in C^r$ and the uniform separation condition (US) holds (infimum of the smallest singular value of $D\Phi_h^{(L)}$ over X is bounded away from zero), $\Phi_h^{(L)}$ is bi-Lipschitz. The empirical delay vectors then constitute a second witness: they certify that points are being separated by the dynamics. The Dynamics Theorem shows that this kernel witness converges to the true separation at rate $n^{-\beta/(2\beta+d)}$ and names the places where it can fail.

The Conjunction Theorem (Section 7). The two witnesses measure the same event. For deterministic T, the TV separation $d_L(x, x') = 1$ if and only if x and x' are separated by $\mathcal{O}_h^{(L)}$ — not merely correlated, but identical. This algebraic identity ($\sigma(\Phi_h^{(L)}) = \mathcal{O}_h^{(L)}$) is the Markov bridge. Under reconstruction and (US), both witnesses fire correctly and certify the same object. The theorem also names two failure modes: reconstruction failure makes the algebra witness fail; violation of (US) makes the kernel witness fail even when reconstruction holds.

The σ-algebra approximation error $\delta(L)$ is observable without knowing the mixing rate, the dimension d, or the smoothness s. The elbow rule for $\hat{\delta}(L, n)$ operates upstream of the rate: it asks whether the algebra has captured $\mathcal{B}$, not at what rate the approximation is converging. For deterministic T, all three witnesses certify the same object, and finite data suffices to detect it.

2 Setup and Notation

2.1 The measure-preserving system

Let $(X, \mathcal{B}, \mu)$ be a probability space where X is a compact smooth d-dimensional Riemannian manifold and μ is a Borel probability measure with smooth, everywhere-positive density $\rho : X \to \mathbb{R}_{>0}$ satisfying

$$\rho_{\min} \;:=\; \operatorname{essinf}_X \rho \;>\; 0.$$

Let $T : X \to X$ be a μ-preserving measurable map. We make no injectivity or invertibility assumption on T beyond what is stated in each hypothesis.

2.2 The observable and the delay map

Fix an observable $h \in L^\infty(\mu)$ with $\|h\|_{L^\infty} =: a < \infty$. For each lag depth $L \geq 0$ define the *delay map*

$$\Phi_h^{(L)} : X \to \mathbb{R}^{L+1}, \qquad \Phi_h^{(L)}(x) \;=\; \big(h(x),\, h(Tx),\, \ldots,\, h(T^L x)\big).$$

The *observable σ-algebra at lag L* is

$$\mathcal{O}_h^{(L)} \;:=\; \sigma(\Phi_h^{(L)}) \;=\; \sigma\big\{h \circ T^k : 0 \leq k \leq L\big\}.$$

The *delay-polynomial class* of degree $D \geq 1$ is

$$\mathcal{A}_h^{(L)} \;:=\; \big\{p \circ \Phi_h^{(L)} : p \text{ is a polynomial of degree} \leq D \text{ on } \mathbb{R}^{L+1}\big\} \;\subseteq\; L^2(X, \mathcal{O}_h^{(L)}, \mu).$$

Since $\Phi_h^{(L)}(X) \subseteq [-a, a]^{L+1}$, all elements of $\mathcal{A}_h^{(L)}$ are bounded.

2.3 The approximation error

The central quantity is the σ-algebra approximation error

$$\delta(L) \;:=\; \delta(\mathcal{O}_h^{(L)}) \;=\; \sup_{S \in \mathcal{B}} \inf_{E \in \mathcal{O}_h^{(L)}} \mu(S \triangle E),$$

which measures how much of $\mathcal{B}$ remains outside $\mathcal{O}_h^{(L)}$. It is non-increasing in L, vanishes if and only if reconstruction holds, and under (G1)–(G2) on a smooth manifold reaches zero by lag $L_0 \leq d-1$ [18].

2.4 Fibre structure and collision probability

The delay map $\Phi_h^{(L)}$ partitions X into *fibres*: for each $z \in \mathbb{R}^{L+1}$, define

$$F_z := (\Phi_h^{(L)})^{-1}(z), \qquad p_z := \mu(F_z), \qquad \mu_z := \mu(\cdot \mid F_z).$$

Call z an *ε-large fibre* if $p_z \geq \varepsilon$. The push-forward $\nu_L := (\Phi_h^{(L)})_*\mu$ is the distribution of delay vectors.

The set of unseparated pairs $R_L := \{(x, x') : \Phi_h^{(L)}(x) = \Phi_h^{(L)}(x')\}$ decomposes as $R_L = \bigsqcup_z F_z \times F_z$, giving

$$(\mu \otimes \mu)(R_L) = \int p_z^2 \, d\nu_L(z). \tag{1}$$

This quantity is the *collision probability* of the delay-vector distribution: it is the probability that two independent draws from μ produce the same delay vector.

2.5 The data model

We observe a time series $x_1, x_2, \ldots, x_n$ which is an orbit segment of T: $x_i = T^{i-1}x_1$ for some initial condition $x_1 \in X$ drawn from μ. The empirical measure is $\mu_n := \frac{1}{n}\sum_{i=1}^n \delta_{x_i}$. We do not assume ergodicity or density of the orbit; each result states the mixing hypothesis it uses.

2.6 Structural hypotheses

The following hypotheses are referenced throughout. They are not assumed globally; each theorem states exactly which subset it uses.

(G1) *No short periodic orbits:* T has no periodic orbit of period $\leq L$ in $\operatorname{supp}(\mu)$. This ensures the orbit points $x, Tx, \ldots, T^L x$ are distinct μ-a.e., which is used in the Sard–Smale argument for full rank of $\Phi_h^{(L)}$.

(G2) *Non-degenerate observable:* $h \in C^2(X)$ with $dh \neq 0$ μ-a.e. This places h in the Sard–Smale regular-value residual set and ensures $\Phi_h^{(L)}$ has rank $\min(L+1, d)$ generically.

(US) *Uniform separation:* $\inf_{x \in X} \sigma_{\min}(D\Phi_h^{(L)}|_x) > 0$, where $\sigma_{\min}$ denotes the smallest singular value. This is a uniform lower Lipschitz bound on $\Phi_h^{(L)}$, strictly stronger than (G1)–(G2) which give full rank only μ-a.e. Condition (US) is used only in the Dynamics and Conjunction theorems (Sections 6 and 7).

2.7 The empirical witness

The empirical analogue of $\delta(L)$ is

$$\hat{\delta}(L, n) := \sup_{p \in \mathcal{P}_D^{(L)},\, \|p\|_{L^2(\mu)} \leq 1} \left\| p - \hat{\mathbb{E}}[p \mid \Phi_h^{(L)}] \right\|_{L^2(\mu_n)}^2,$$

where $\mathcal{P}_D^{(L)}$ is the class of degree-D polynomials in the delay coordinates $\Phi_h^{(L)}(x)$ (normalised in $L^2(\mu)$), and $\hat{\mathbb{E}}[p \mid \Phi_h^{(L)}]$ is the Nadaraya–Watson kernel regression estimator (Definition 4.5).

The main result of Section 4 is that $\hat{\delta}(L, n)$ concentrates around $\delta(L)$ at rate $\varepsilon_n = C\, n^{-2s/(2s+d)}$ under exponential mixing, provided $d > 2s$.

3 Bias Bound and Conditional Variance

Let $(X, \mathcal{B}, \mu)$ be a probability space and $\mathfrak{m} \subseteq \mathcal{B}$ a sub-σ-algebra. Define the *σ-algebra approximation error*

$$\delta(\mathfrak{m}) \;:=\; \sup_{S\in\mathcal{B}} \inf_{E\in\mathfrak{m}} \mu(S \triangle E). \tag{2}$$

This measures how well $\mathfrak{m}$ approximates $\mathcal{B}$: it is zero if and only if $\mathfrak{m} = \mathcal{B}$ modulo μ-null sets.

Lemma 3.1 (Conditional variance identity)**.** *For any $S \in \mathcal{B}$:*

$$\mathbb{E}\big[\mathrm{Var}(\mathbb{1}_S \mid \mathcal{O}_h^{(L)})\big] \;=\; \frac{1}{2}\int_{R_L} \big|\mathbb{1}_S(x) - \mathbb{1}_S(x')\big|^2 \, d(\mu\otimes\mu)(x, x').$$

Proof. By disintegration, $(\mu\otimes\mu)|_{R_L} = \int \mu_z \otimes \mu_z \, d\nu_L(z)$. On each fibre F_z:

$$\tfrac{1}{2}\int_{F_z\times F_z} |\mathbb{1}_S(x) - \mathbb{1}_S(x')|^2 \, d(\mu_z\otimes\mu_z) \;=\; \mu_z(S)\,\mu_z(S^c).$$

Integrating with weight p_z: $\frac{1}{2}\int_{R_L} |\mathbb{1}_S - \mathbb{1}_{S'}|^2 \, d(\mu\otimes\mu) = \int p_z\, \mu_z(S)\mu_z(S^c)\, d\nu_L(z)$. Since $\mathrm{Var}(\mathbb{1}_S \mid \mathcal{O}_h^{(L)})(x) = \mu_z(S)\mu_z(S^c)$ on F_z, both sides equal $\int \mu_z(S)\mu_z(S^c)\, d\nu_L(z)$. □

Corollary 3.2 (Easy direction)**.** $\delta(L) \;\le\; \frac{1}{2}\,(\mu\otimes\mu)(R_L)$.

Proof. In Lemma 3.1, bound $|\mathbb{1}_S(x) - \mathbb{1}_S(x')|^2 \;\le\; 1$ and take the supremum over S using $\delta(L) = \sup_S \mathbb{E}[\mathrm{Var}(\mathbb{1}_S \mid \mathcal{O}_h^{(L)})]$. □

Remark 3.3 (Finite analogue of collective exhaustion)**.** Corollary 3.2 and its converse (Corollary 5.12, proved under fibre mixing in §5) together say: algebraic indistinguishability $\delta(L)$ vanishes if and only if the mass of indistinguishable pairs $(\mu\otimes\mu)(R_L)$ vanishes. This is the finite-sample counterpart of collective exhaustion [19]. There, CE asserts that set-theoretic vanishing must be witnessed at some finite level of the refinement order. Here, the analogous principle is that observational indistinguishability cannot persist globally without manifesting as a non-vanishing mass of indistinguishable pairs at finite lag L. The fibre mixing condition plays an analogous structural role to CE: it is a sufficient condition under which the global and pairwise witnesses are comparable. Whether fibre mixing is itself derivable from dynamical conditions — or is genuinely irreducible, as CE is proved to be in [19] — remains open; see the open problems in Section 8.

Lemma 3.4 (Bias bound)**.** *For every $f \in L^\infty(\mu)$,*

$$\big\|f - \mathbb{E}[f \mid \mathfrak{m}]\big\|_{L^2(\mu)} \;\le\; 2\|f\|_{L^\infty} \cdot \delta(\mathfrak{m})^{1/2}.$$

Proof. Step 1 (Indicators). Fix $S \in \mathcal{B}$. By definition of $\delta(\mathfrak{m})$, there exists $E \in \mathfrak{m}$ with $\mu(S\triangle E) \le \delta(\mathfrak{m})$. Since $\mathbb{1}_E \in L^2(X, \mathfrak{m}, \mu)$ and $\mathbb{E}[\mathbb{1}_S \mid \mathfrak{m}]$ is the L^2-best $\mathfrak{m}$-measurable approximation to $\mathbb{1}_S$,

$$\|\mathbb{1}_S - \mathbb{E}[\mathbb{1}_S \mid \mathfrak{m}]\|_{L^2}^2 \;\le\; \|\mathbb{1}_S - \mathbb{1}_E\|_{L^2}^2 \;=\; \mu(S\triangle E) \;\le\; \delta(\mathfrak{m}).$$

Step 2 (Bounded f via layer cake). For $f \in L^\infty(\mu)$ with $\|f\|_{L^\infty} \le M$, the layer cake representation gives $f = \int_{-M}^{M} \mathbb{1}_{S_t}\, dt$ where $S_t = \{x : f(x) > t\}$. Since $\mathbb{E}[\cdot \mid \mathfrak{m}]$ is linear and continuous in L^2,

$$f - \mathbb{E}[f \mid \mathfrak{m}] \;=\; \int_{-M}^{M} \big(\mathbb{1}_{S_t} - \mathbb{E}[\mathbb{1}_{S_t} \mid \mathfrak{m}]\big)\, dt.$$

Applying Minkowski's inequality for integrals and Step 1:

$$\big\|f - \mathbb{E}[f \mid \mathfrak{m}]\big\|_{L^2} \;\le\; \int_{-M}^{M} \big\|\mathbb{1}_{S_t} - \mathbb{E}[\mathbb{1}_{S_t} \mid \mathfrak{m}]\big\|_{L^2}\, dt \;\le\; 2M \cdot \delta(\mathfrak{m})^{1/2}.$$ □

We now specialise to the delay setting. Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system and $h \in L^\infty(\mu)$. Write $\mathcal{O}_h^{(L)} = \sigma(\Phi_h^{(L)})$ for the observable σ-algebra at lag L, where $\Phi_h^{(L)}(x) = (h(x), h(Tx), \ldots, h(T^L x))$, and $\mathcal{A}_h^{(L)}$ for the class of degree-D polynomials in the delay coordinates $\Phi_h^{(L)}(x)$. Let $\hat{f}_n^{(L)}$ denote the empirical risk minimiser over $\mathcal{A}_h^{(L)}$.

Lemma 3.5 (Three-term decomposition)**.** *For any $f \in L^\infty(\mu)$, any $L \geq 0$, and any $D \geq 1$:*

$$\|f - \hat{f}_n^{(L)}\|_{L^2(\mu)} \leq \underbrace{2\|f\|_{L^\infty} \cdot \delta(L)^{1/2}}_{bias} + \underbrace{\inf_{g \in \mathcal{A}_h^{(L)}} \|\mathbb{E}[f \mid \mathcal{O}_h^{(L)}] - g\|_{L^2(\mu)}}_{approx.} + \underbrace{\|g_D^* - \hat{f}_n^{(L)}\|_{L^2(\mu)}}_{statistical}, \quad (3)$$

where $\delta(L) := \delta(\mathcal{O}_h^{(L)})$ as in (**??**) *and $g_D^* = \operatorname{argmin}_{g \in \mathcal{A}_h^{(L)}} \|\mathbb{E}[f \mid \mathcal{O}_h^{(L)}] - g\|_{L^2(\mu)}$.*

Proof. Triangle inequality applied to the decomposition $f - \hat{f}_n^{(L)} = (f - \mathbb{E}[f \mid \mathcal{O}_h^{(L)}]) + (\mathbb{E}[f \mid \mathcal{O}_h^{(L)}] - g_D^*) + (g_D^* - \hat{f}_n^{(L)})$, with the first term bounded by Lemma 3.4. □

The three-term bound controls the forward error. For the stopping rule to be honest, we also need a reverse bound: when $\delta(L)$ is large, the estimator $\hat{f}_n^{(L)}$ cannot approximate f well, regardless of n.

Lemma 3.6 (Observable reverse lower bound)**.** *There exists a universal constant $c > 0$ such that for any $f \in L^\infty(\mu)$ with $\|f\|_{L^\infty} \geq \frac{1}{2}$ and any $L \geq 0$:*

$$\|\mathbb{1}_S - \hat{f}_n^{(L)}\|_{L^2(\mu_n)} \geq \frac{\delta(L)^{1/2}}{2} - C' \cdot n^{-s/(2s+d)}$$

with probability tending to 1 as $n \to \infty$, where $S \in \mathcal{B}$ is the set achieving the supremum in (**??**) *at lag L and $\mu_n = \frac{1}{n}\sum_{i=1}^n \delta_{x_i}$ is the empirical measure.*

Proof. Step A (Population CE gap). Let $S^* \in \mathcal{B}$ achieve the supremum in $\delta(L) = \sup_S \inf_{E \in \mathcal{O}_h^{(L)}} \mu(S \triangle E)$. For any $E \in \mathcal{O}_h^{(L)}$:

$$\|\mathbb{1}_{S^*} - \mathbb{E}[\mathbb{1}_{S^*} \mid \mathcal{O}_h^{(L)}]\|_{L^2(\mu)}^2 \leq \|\mathbb{1}_{S^*} - \mathbb{1}_E\|_{L^2(\mu)}^2 = \mu(S^* \triangle E) \geq \delta(L) - \varepsilon$$

for any $\varepsilon > 0$. Taking $\varepsilon \to 0$: $\|\mathbb{1}_{S^*} - \mathbb{E}[\mathbb{1}_{S^*} \mid \mathcal{O}_h^{(L)}]\|_{L^2(\mu)} \geq \delta(L)^{1/2}$.

Since the conditional expectation is the L^2-projection, $\|\mathbb{1}_{S^*} - g\|_{L^2(\mu)} \geq \delta(L)^{1/2}$ for all $g \in \mathcal{A}_h^{(L)} \subseteq L^2(X, \mathcal{O}_h^{(L)}, \mu)$.

Step B (Empirical vs population norm). By Hoeffding's inequality and the bounded difference property of μ_n, $\|\mathbb{1}_{S^*} - g\|_{L^2(\mu_n)} \geq \|\mathbb{1}_{S^*} - g\|_{L^2(\mu)} - O(n^{-1/2})$ with high probability, uniformly over $g \in \mathcal{A}_h^{(L)}$.

Step C (Empirical norm vs $\hat{f}_n^{(L)}$). Since $\hat{f}_n^{(L)}$ minimises empirical L^2 error over $\mathcal{A}_h^{(L)}$, and $\|\mathbb{1}_{S^*} - g\|_{L^2(\mu_n)} \geq \delta(L)^{1/2} - O(n^{-1/2})$ for all $g \in \mathcal{A}_h^{(L)}$, we have $\|\mathbb{1}_{S^*} - \hat{f}_n^{(L)}\|_{L^2(\mu_n)} \geq \frac{\delta(L)^{1/2}}{2} - C' n^{-s/(2s+d)}$ with high probability, absorbing the $O(n^{-1/2})$ into the statistical rate. □

4 Concentration of the Empirical Witness

Throughout this section: (T, h) satisfies (G1)–(G2), X is a compact smooth d-manifold, μ has smooth positive density, and $d > 2s$.

4.1 Uniform boundedness

Let $\mathcal{P}_D^{(L)}$ denote the class of degree-D polynomials in the delay coordinates $\Phi_h^{(L)}(x) = (h(x), \dots, h(T^L x))$, normalised so $\|p\|_{L^2(\mu)} \le 1$. Since $h \in L^\infty$ with $\|h\|_{L^\infty} =: a < \infty$, the delay coordinates lie in the compact box $[-a, a]^{L+1}$.

Lemma 4.1 (Uniform boundedness). *Define*

$$M_{D,L} \;:=\; \binom{D+L+1}{L+1}^{1/2} \cdot (2D+1)^{1/2} \cdot (2a)^{-(L+1)/2} \cdot \rho_{\min}^{-1/2},$$

where $\rho_{\min} = \operatorname{essinf}_X \rho > 0$ *is the infimum of the density of* μ*. Then for every* $p \in \mathcal{P}_D^{(L)}$*:*

$$\|p\|_{L^\infty} \;\le\; M_{D,L} \cdot \|p\|_{L^2(\mu)}.$$

Proof. Rescale to the unit box by setting $y_i = x_i/a$; polynomials of degree $\le D$ in $x \in [-a,a]^{L+1}$ correspond to polynomials $\tilde{p}$ of degree $\le D$ in $y \in [-1,1]^{L+1}$. Expand $\tilde{p}$ in the tensor-product Legendre basis $\{P_\alpha\}_{|\alpha| \le D}$, where $P_\alpha(y) = \prod_{i=1}^{L+1} P_{\alpha_i}(y_i)$ and $\|P_k\|_{L^\infty[-1,1]} = 1$. Since $\|P_\alpha\|_{L^\infty} = 1$, Cauchy–Schwarz gives

$$|\tilde{p}(y)| \;\le\; \sum_{|\alpha| \le D} |c_\alpha| \;\le\; N(D, L+1)^{1/2} \Big(\sum_{|\alpha| \le D} |c_\alpha|^2 \Big)^{1/2},$$

where $N(D, L{+}1) = \binom{D+L+1}{L+1}$ counts the basis elements. From the L^2 orthogonality $\|P_\alpha\|^2_{L^2[-1,1]} = \prod_i (2\alpha_i + 1)^{-1}$, the coefficient norm satisfies

$$\sum_{|\alpha| \le D} |c_\alpha|^2 \;\le\; (2D+1)\, \|\tilde{p}\|^2_{L^2},$$

since the maximum of $\prod_i (2\alpha_i + 1)$ over $|\alpha| \le D$ is $2D+1$, achieved at $\alpha = (D, 0, \dots, 0)$ — the constraint $|\alpha| \le D$ forces all weight onto one coordinate, giving an L-independent maximum. Converting back via $\|\tilde{p}\|_{L^2[-1,1]^{L+1}} = (2a)^{-(L+1)/2} \|p\|_{L^2(\mathrm{vol})}$ and $\|p\|_{L^2(\mathrm{vol})} \le \rho_{\min}^{-1/2} \|p\|_{L^2(\mu)}$ yields the result. The subsequent arguments require $\|h\|_{L^\infty} \ge \frac{1}{2}$, equivalently $2a \ge 1$, so that $M_{D,L}$ grows at most polynomially in L. $\square$

4.2 Concentration around the mean

Define the function class

$$\mathcal{F}_{L,D} \;=\; \Big\{ x \mapsto \big(p(x) - \hat{\mathbb{E}}[p \mid \Phi_h^{(L)}](x)\big)^2 \;:\; p \in \mathcal{P}_D^{(L)},\ \|p\|_{L^2(\mu)} \le 1 \Big\},$$

where $\hat{\mathbb{E}}[p \mid \Phi_h^{(L)}]$ is the Nadaraya–Watson estimator (Definition 4.5). Then $\hat{\delta}(L, n) = \sup_{f \in \mathcal{F}_{L,D}} \frac{1}{n} \sum_{i=1}^n f(x_i)$.

Proposition 4.2 (Concentration, $\star$). *Assume $d > 2s$. There exist constants $c, C > 0$ depending only on $(d, s, \|h\|_{L^\infty})$ such that for all $t > 0$ and all $n \ge 1$:*

$$P\big(|\hat{\delta}(L,n) - \mathbb{E}[\hat{\delta}(L,n)]| > t\big) \;\le\; 2 \exp\Big(-\frac{c\, n\, t^2}{M_{D,L}^4 \cdot D^d} \Big). \tag{$\star$}$$

In particular, setting $t = \varepsilon_n/2$ with $\varepsilon_n = C'\, n^{-2s/(2s+d)}$:

$$P\big(|\hat{\delta}(L,n) - \mathbb{E}[\hat{\delta}(L,n)]| > \varepsilon_n/2\big) \;\le\; 2 \exp\Big(-c'\, n^{(d-2s)/(2s+d)} \Big) \;\to\; 0.$$

Proof. Step 1 (Boundedness). By Lemma 4.1, every $p \in \mathcal{P}_D^{(L)}$ with $\|p\|_{L^2(\mu)} \le 1$ satisfies $\|p\|_{L^\infty} \le M_{D,L}$. Hence every $f = (p - \hat{\mathbb{E}}[p|\Phi_h^{(L)}])^2 \in \mathcal{F}_{L,D}$ satisfies $\|f\|_{L^\infty} \le 4M_{D,L}^2 =: B_{L,D}^2$.

Step 2 (McDiarmid). Replacing one observation x_i with x_i' changes $\hat{\delta}(L,n)$ by at most $B_{L,D}^2/n$. By the McDiarmid bounded-difference inequality:

$$P\big(|\hat{\delta}(L,n) - \mathbb{E}[\hat{\delta}(L,n)]| > t\big) \;\le\; 2\exp\Big(-\frac{2n^2t^2}{n\cdot(B_{L,D}^2/n)^2}\Big) \;=\; 2\exp\Big(-\frac{n\,t^2}{2B_{L,D}^4/n}\Big).$$

Step 3 (Rademacher bias). By the symmetrisation inequality [2]:

$$\mathbb{E}[\hat{\delta}(L,n)] - \delta(L) \;\le\; 2\,\mathfrak{R}_n(\mathcal{F}_{L,D}).$$

By the Ledoux–Talagrand contraction lemma, since $f = (p - \cdot)^2$ is $2B_{L,D}$-Lipschitz in the second argument:

$$\mathfrak{R}_n(\mathcal{F}_{L,D}) \;\le\; 2B_{L,D}\,\mathfrak{R}_n(\mathcal{P}_D^{(L)}).$$

In the Takens regime the image $\Phi_h^{(L)}(X)$ is a d-dimensional submanifold, so the VC dimension of $\mathcal{P}_D^{(L)}$ restricted to $\Phi_h^{(L)}(X)$ is $O(D^d)$ [9]. Hence $\mathfrak{R}_n(\mathcal{P}_D^{(L)}) \le C\sqrt{D^d/n}$, giving $\mathfrak{R}_n(\mathcal{F}_{L,D}) \le C\,M_{D,L}\sqrt{D^d/n}$.

Step 4 (Combined). Combining Steps 2–3 and absorbing constants:

$$P\big(|\hat{\delta}(L,n) - \delta(L)| > t\big) \;\le\; 2\exp\Big(-\frac{c\,n\,t^2}{M_{D,L}^4\,D^d}\Big).$$

Setting $t = \varepsilon_n/2 = C'n^{-2s/(2s+d)}/2$, the exponent becomes

$$c\,n\,\varepsilon_n^2/(M_{D,L}^4\,D^d) \;\propto\; n^{1-4s/(2s+d)} \;=\; n^{(d-2s)/(2s+d)}.$$

This tends to $+\infty$ if and only if $d > 2s$, proving the final claim. □

Remark 4.3 (The $d > 2s$ hypothesis)**.** When $d \le 2s$, the function class $\mathcal{F}_{L,D}$ is too rich for n samples to resolve the fine structure of $\delta(L)$ at the required scale. The fix via localised Rademacher complexity [1] is technically available but left to future work; we take $d > 2s$ as a standing hypothesis.

4.3 Source 1: the LLN gap

Lemma 4.4 (LLN gap)**.** *With notation as above:*

$$\sup_{p\in\mathcal{P}_D^{(L)}} \Big| \|p - \mathbb{E}[p \mid \Phi_h^{(L)}]\|_{L^2(\mu_n)}^2 - \|p - \mathbb{E}[p \mid \Phi_h^{(L)}]\|_{L^2(\mu)}^2 \Big| \;\le\; C\sqrt{D^d/n}$$

almost surely as $n \to \infty$, where C depends on $(d, D, M_{D,L})$.

Proof. The class $\{(p - \mathbb{E}[p|\Phi_h^{(L)}])^2 : p \in \mathcal{P}_D^{(L)}\}$ is a VC class of dimension $O(D^d)$. The uniform Glivenko–Cantelli theorem [9] gives a.s. convergence of the empirical mean to the population mean, at rate $O(\sqrt{D^d/n})$. □

4.4 Source 2: estimator bias

The estimator $\hat{\mathbb{E}}[p \mid \Phi_h^{(L)}]$ is the Nadaraya–Watson (NW) kernel regression estimator.

Definition 4.5 (Nadaraya–Watson estimator)**.** For bandwidth $h_n > 0$ and kernel $K : \mathbb{R}^{L+1} \to \mathbb{R}_{\geq 0}$ (compactly supported, symmetric, $\int K = 1$):

$$\hat{\mathbb{E}}[p \mid \Phi_h^{(L)}(x)] \;:=\; \frac{\sum_{j=1}^n K\big(\frac{\Phi_h^{(L)}(x)-\Phi_h^{(L)}(x_j)}{h_n}\big)\, p(x_j)}{\sum_{j=1}^n K\big(\frac{\Phi_h^{(L)}(x)-\Phi_h^{(L)}(x_j)}{h_n}\big)}.$$

Proposition 4.6 (Source 2 estimator bias)**.** *Assume:*

(i) $T, h \in C^r$ with $r \geq 2$, μ smooth positive density ρ with $\rho_{\min} > 0$.

(ii) $\Phi_h^{(L)}$ has full rank μ-a.e. (follows from (G1)–(G2) and Lemma 5.2).

(iii) T is exponentially mixing with rate $\lambda > 0$ and $\|h\|_{L^\infty} \geq \frac{1}{2}$.

(iv) The NW bandwidth satisfies $h_n \sim n^{-1/(2\beta+d_{\mathrm{eff}}(L))}$ with $d_{\mathrm{eff}}(L) = \min(L+1, d)$ and $\beta \geq 1$.

Then there exists $C_2(D, L) > 0$ such that for all $p \in \mathcal{P}_D^{(L)}$:

$$\mathbb{E}\|\hat{\mathbb{E}}[p \mid \Phi_h^{(L)}] - \mathbb{E}[p \mid \Phi_h^{(L)}]\|^2_{L^2(\mu)} \;\leq\; C_2(D, L) \cdot n^{-2\beta/(2\beta+d_{\mathrm{eff}}(L))},$$

uniformly in $p \in \mathcal{P}_D^{(L)}$, where $C_2(D, L) = 2M_{D,L}^2 C_1^2 A^{2\beta}$ and A is a constant depending on $(\beta, d_{\mathrm{eff}}(L), \rho_{\min})$.

Proof. The proof is a four-link chain.

Link 1 (Smooth disintegration). Under conditions (i)–(ii), the Rokhlin disintegration $\mu = \int \mu_z \, d((\Phi_h^{(L)})_*\mu)(z)$ has fibre measures μ_z supported on the smooth fibres $F_z = \{\Phi_h^{(L)} = z\}$. An implicit-function-theorem argument (three sub-steps: transfer to common surface via a C^{r-1} diffeomorphism; bound weight variation; bound Jacobian factor) gives:

$$\|\mu_z - \mu_{z'}\|_{\mathrm{TV}} \;\leq\; C_1 \, |z - z'|^\beta, \qquad \beta = 1 \text{ (Lipschitz floor)},$$

where C_1 depends on $(\|T\|_{C^r}, \|h\|_{C^r}, \rho_{\min})$. (The optimal exponent $\beta = r - 1$ follows from the full Taylor expansion of the fibre weight; the floor $\beta = 1$ is sufficient for convergence.)

Link 2 (Bernstein–Markov). From Link 1 and Lemma 4.1: for $p \in \mathcal{P}_D^{(L)}$ with $\|p\|_{L^2(\mu)} \leq 1$,

$$\big|\mathbb{E}[p \mid \Phi_h^{(L)} = z] - \mathbb{E}[p \mid \Phi_h^{(L)} = z']\big| \;=\; \Big|\int p \, d\mu_z - \int p \, d\mu_{z'}\Big| \;\leq\; M_{D,L} \cdot C_1 \cdot |z - z'|^\beta.$$

Hence $z \mapsto \mathbb{E}[p \mid \Phi_h^{(L)} = z]$ is Hölder(β) with constant $\Lambda_{D,L} := M_{D,L} C_1$, uniformly over $p \in \mathcal{P}_D^{(L)}$.

Link 3 (NW bias-variance balance). Standard NW theory [9] for a Hölder(β) regression function on a $d_{\mathrm{eff}}(L)$-dimensional domain gives:

$$\mathbb{E}\|\hat{\mathbb{E}}[p|\Phi_h^{(L)}] - \mathbb{E}[p|\Phi_h^{(L)}]\|^2_{L^2(\mu)} \;\leq\; \Lambda_{D,L}^2 \, h_n^{2\beta} + \frac{C_K \, M_{D,L}^2}{n \, h_n^{d_{\mathrm{eff}}(L)} \, f_{\min}},$$

where C_K depends on the kernel K and $f_{\min}$ is the minimum of the marginal density of $(\Phi_h^{(L)})_*\mu$. The $M_{D,L}^2$ factor appears in both terms. Balancing: $\Lambda_{D,L}^2 h_n^{2\beta} = C_K M_{D,L}^2/(n h_n^{d_{\mathrm{eff}}(L)} f_{\min})$ gives $h_n^* \sim A \cdot n^{-1/(2\beta+d_{\mathrm{eff}}(L))}$ where A depends only on $(\beta, d_{\mathrm{eff}}(L), C_K, f_{\min}, C_1)$ — not on D, L, or p. The $M_{D,L}^2$ factors cancel from both sides of the balance equation.

Link 4 (Uniformity). Substituting h_n^*, both terms contribute $C_2(D, L) \cdot n^{-2\beta/(2\beta+d_{\mathrm{eff}}(L))}$ with $C_2(D, L) = 2M_{D,L}^2 C_1^2 A^{2\beta}$. Since the bound is p-free, it holds uniformly over $\mathcal{P}_D^{(L)}$. □

Theorem 4.7 (Full convergence $\hat{\delta}(L,n) \to \delta(L)$)**.** *Under the hypotheses of Proposition 4.6 and with $d > 2s$:*

$$\mathbb{E}\big|\hat{\delta}(L,n) - \delta(L)\big| \;\le\; C\sqrt{D^d/n} \;+\; C_2(D,L)^{1/2} \cdot n^{-\beta/(2\beta+d_{\mathrm{eff}}(L))}$$

and $P(|\hat{\delta}(L,n) - \delta(L)| > \varepsilon_n) \to 0$ exponentially, where $\varepsilon_n = C' n^{-2s/(2s+d)}$.

Proof. Decompose:

$$|\hat{\delta}(L,n) - \delta(L)| \;\le\; \underbrace{|\hat{\delta}(L,n) - \mathbb{E}[\hat{\delta}(L,n)]|}_{\text{concentration, } (\star)} \;+\; \underbrace{|\mathbb{E}[\hat{\delta}(L,n)] - \delta(L)|}_{\text{bias}}$$

The bias splits as Source 1 + Source 2: Source 1 is bounded by $C\sqrt{D^d/n}$ (Lemma 4.4); Source 2 is bounded by $C_2(D,L)^{1/2} \cdot n^{-\beta/(2\beta+d_{\mathrm{eff}}(L))}$ (Proposition 4.6). The concentration term is bounded by $(\star)$ (Proposition 4.2). Combining at $t = \varepsilon_n/2$ gives the probability statement. □

5 The Algebra Theorem

This section assembles the results of Sections 3 and 4 into the Algebra Theorem: the data-driven stopping rule $\hat{L}^*$ achieves the minimax-optimal rate $n^{-s/(2s+d)}$ without knowing the oracle lag $L^*(n)$ or the mixing rate of T.

5.1 Effective dimension and the piecewise rate

Definition 5.1 (Effective dimension)**.** The *effective dimension at lag L* is

$$d_{\mathrm{eff}}(L) \;:=\; \mathrm{rank}(D\Phi_h^{(L)}|_x) \quad \text{at a generic point } x \in X.$$

(The rank is constant on an open dense set by the rank theorem, so this is well-defined μ-a.e.)

Lemma 5.2 (d_{eff} step function)**.** *Under (G1)–(G2) with X a compact smooth d-manifold and μ smooth positive density:*

$$d_{\mathrm{eff}}(L) \;=\; \min(L+1,\, d) \quad \textit{for all } L \ge 0.$$

Proof. The upper bound $d_{\mathrm{eff}}(L) \le \min(L+1, d)$ holds because $D\Phi_h^{(L)}|_x$ is a linear map from T_xX (dimension d) to $\mathbb{R}^{L+1}$.

For the lower bound we use the Sard–Smale theorem (Smale [23] framework) applied to two failure maps.

Case $L+1 \le d$. Rank $< L+1$ at x means the cotangent vectors $\omega_j(x) := (DT^j|_x)^*(dh|_{T^jx})$ are linearly dependent. Define the failure map

$$\Psi : C^2(X) \times X \times \mathbb{P}^L \to T^*X, \quad (h, x, [\lambda]) \;\mapsto\; d\Big(\textstyle\sum_j \lambda_j\, h \circ T^j\Big)\,\Big|_x.$$

By (G1), the orbit points T^jx are distinct μ-a.e., so bump functions at T^jx can prescribe any cotangent vector at x independently. Hence $D_h\Psi$ is surjective onto T_x^*X (dimension d), and $\Psi^{-1}(0)$ is a Banach submanifold of codimension d. The projection to $C^2(X)$ is Fredholm of index $d + L - d = L$, so by Sard–Smale the regular values form a residual set $R \subset C^2(X)$. For $h \in R$, the failure locus in X has dimension $\le L < d$, hence μ-measure zero.

By (G2), $dh \neq 0$ μ-a.e., which forces every critical point of $F_\lambda = \sum_j \lambda_j h \circ T^j$ to lie in the codimension-d submanifold $S_\lambda = \{dh_x \in \mathrm{span}\{\omega_1(x), \ldots, \omega_L(x)\}\}$ (a finite set for each $[\lambda]$, since X is d-dimensional). Hence h is a regular value of π, i.e., $h \in R$.

Case $L+1>d$. Immersion failure requires $\exists\, v \neq 0$ with $D\Phi_h^{(L)}|_x \cdot v = 0$. Define

$$\Xi : C^2(X) \times (TX \setminus 0) \to \mathbb{R}^{L+1}, \quad (h,(x,v)) \;\mapsto\; \big(dh_{T^j x}(DT^j|_x \cdot v)\big)_{j=0}^{L}.$$

By (G1), $D_h\Xi$ is surjective onto $\mathbb{R}^{L+1}$, so $\Xi^{-1}(0)$ has codimension $L+1$ and the projection to $C^2(X)$ is Fredholm of index $2d-1-(L+1) = 2d-L-2$. For $L \geq d$: $2d-L-2 \leq d-2 < d-1$, so the failure locus in X has dimension $< d-1$ and μ-measure zero. Condition (G2) places h in the regular-value set by the same argument as Case 1. ☐

Proposition 5.3 (Piecewise pre-Takens rate)**.** *Assume (G1)–(G2), X compact smooth d-manifold, μ smooth positive density, $f \in$ Hölder$(s) \cap L^\infty(\mu)$, $d > 2s$, and $T \in C^r$ with $r \geq 2$. Let $\hat{f}_n^{(L)}$ be the empirical risk minimiser over $\mathcal{A}_h^{(L)}$ with $D = D^*(n,L)$ chosen to balance approximation and statistical terms. Then for each $L \geq 0$:*

$$\big\|f - \hat{f}_n^{(L)}\big\|_{L^2(\mu)} \;\leq\; 2\|f\|_{L^\infty} \cdot \delta(L)^{1/2} \;+\; C(f,d,s) \cdot n^{-s/(2s+d_{\text{eff}}(L))}$$

with probability tending to 1, where $d_{\text{eff}}(L) = \min(L+1,d)$. In the Takens regime ($L \geq d-1$): $d_{\text{eff}}(L) = d$ and the rate locks at $n^{-s/(2s+d)}$.

Proof. Apply the three-term decomposition (Lemma 3.5). The bias term is bounded by $2\|f\|_{L^\infty} \cdot \delta(L)^{1/2}$ (Lemma 3.4). By Lemma 5.2, $\Phi_h^{(L)}(X)$ is a $d_{\text{eff}}(L)$-dimensional smooth submanifold. On this submanifold, $\mathbb{E}[f \mid \mathcal{O}_h^{(L)}](x) = g^*(\Phi_h^{(L)}(x))$ for some $g^* \in$ Hölder(s); standard polynomial approximation on a $d_{\text{eff}}(L)$-dimensional smooth domain gives approximation error $\leq C_{\text{approx}} \cdot D^{-s/d_{\text{eff}}(L)}$. The statistical term is $\leq C_{\text{stat}}(D^{d_{\text{eff}}(L)}/n)^{1/2}$ by Proposition 4.2 with d replaced by $d_{\text{eff}}(L)$. Balancing: $D^* \sim n^{d_{\text{eff}}(L)/(2s+d_{\text{eff}}(L))}$ and both terms are $O(n^{-s/(2s+d_{\text{eff}}(L))})$. ☐

5.2 The oracle lag and the elbow stopping rule

Definition 5.4 (Oracle lag)**.** The *oracle lag* is

$$L^*(n) \;:=\; \min\big\{L \geq 0 : \delta(L) \leq C_0 \cdot n^{-2s/(2s+d)}\big\},$$

where $C_0 > 0$ is a constant depending on $\|f\|_{L^\infty}$. At $L = L^*(n)$, the bias term $2\|f\|_{L^\infty} \cdot \delta(L^*)^{1/2}$ is $O(n^{-s/(2s+d)})$, matching the statistical floor.

The oracle lag $L^*(n)$ requires knowledge of the mixing rate, which is unobservable. The empirical witness $\hat{\delta}(L,n)$ sidesteps this: it measures reconstruction quality directly from data.

Definition 5.5 (Elbow stopping rule)**.** Fix a tolerance sequence $\tau_n > 0$. The *elbow stopping rule* is

$$\hat{L}^* \;:=\; \min\big\{L \geq 0 : \hat{\delta}(L+1,n) - \hat{\delta}(L,n) > -\tau_n\big\}.$$

Theorem 5.6 (Elbow location)**.** *Assume the hypotheses of Proposition 5.3 and exponential mixing $\delta(L) \leq Ae^{-\lambda L}$ with resolvability constant $C_\lambda := (1-e^{-\lambda})Ae^{\lambda}/C' > 4$. Set $\tau_n = \frac{C_\lambda}{2} \cdot C' \cdot n^{-2s/(2s+d)}$. Then*

$$P\big(\hat{L}^* = L^*(n)\big) \;\to\; 1 \quad \textit{exponentially in } n.$$

Proof. We show the stopping criterion fires at $L = L^*(n)$ (no overshoot) and not before (no early stop).

No overshoot: $\hat{L}^* \leq L^*(n)$. At $L = L^*(n)$, the population increment satisfies $|\delta(L^*(n)+1) - \delta(L^*(n))| \leq \delta(L^*(n)) \leq C_0 n^{-2s/(2s+d)}$ (non-increasing δ). By Theorem 4.7, $|\hat{\delta}(L,n) - \delta(L)| \leq \varepsilon_n = C' n^{-2s/(2s+d)}$ whp. The empirical increment satisfies $\hat{\delta}(L^*+1,n) - \hat{\delta}(L^*,n) \geq -C_0 n^{-2s/(2s+d)} - 2\varepsilon_n > -\tau_n$ whp, so the criterion fires.

No early stop: $\hat{L}^* \geq L^*(n)$. For $L < L^*(n)$, exponential mixing gives population decrement $\delta(L) - \delta(L+1) \geq \Delta(L) \geq C_\lambda \cdot C' n^{-2s/(2s+d)}$. Tracking via Theorem 4.7:

$$\hat{\delta}(L,n) - \hat{\delta}(L+1,n) \;\geq\; C_\lambda C' n^{-2s/(2s+d)} - 2\varepsilon_n \;=\; (C_\lambda - 2) C' n^{-2s/(2s+d)}.$$

Since $C_\lambda > 4$: $(C_\lambda - 2)C' > \tau_n \cdot 2/C_\lambda \cdot C_\lambda = 2\tau_n / C_\lambda \cdot C_\lambda$, and specifically $(C_\lambda - 2)C' n^{-2s/(2s+d)} \geq \tau_n$ when $C_\lambda - 2 \geq C_\lambda/2$, i.e., $C_\lambda \geq 4$. So the stopping criterion does *not* fire for $L < L^*(n)$, and $\hat{L}^* \geq L^*(n)$.

Remark 5.7 (Scope of the elbow theorem)**.** The resolvability condition $C_\lambda > 4$ is not an artefact: when $C_\lambda \leq 4$ the elbow drop at $L^*(n)$ merges with the noise floor ε_n and the two are indistinguishable from data. For polynomial mixing, $\Delta(L^*(n)) \sim \varepsilon_n / L^*(n) \to 0$ relative to ε_n, so the elbow drops vanish asymptotically and the theorem fails. A separate argument (e.g. integrated elbow) is needed for that case and is not proved here.

□

Theorem 5.8 (Algebra Theorem)**.** *Assume:*

(i) Hypotheses of Proposition 5.3: (G1)–(G2), X compact smooth d-manifold, μ smooth positive density, $f \in$ Hölder$(s) \cap L^\infty(\mu)$*, $d > 2s$, $T \in C^r$, $r \geq 2$.*

(ii) Exponential mixing: $\delta(L) \leq A e^{-\lambda L}$ for some $A, \lambda > 0$.

(iii) Resolvability: $C_\lambda > 4$.

(iv) Exponential mixing for the estimator bias (Proposition 4.6) and $\|h\|_{L^\infty} \geq \frac{1}{2}$.

Let $\hat{L}^*$ be the elbow stopping rule (Definition 5.5) with $\tau_n = \frac{C_\lambda}{2} C' n^{-2s/(2s+d)}$. Then

$$\big\|f - \hat{f}_n^{(\hat{L}^*)}\big\|_{L^2(\mu)} \;=\; O_p\big(n^{-s/(2s+d)}\big).$$

The rate $n^{-s/(2s+d)}$ is minimax-optimal for Hölder(s) *functions on a d-dimensional domain. No oracle input (mixing rate, $L^*(n)$, Sobolev index s) is required.*

Proof. On the event $\{\hat{L}^* = L^*(n)\}$, which has probability tending to 1 by Theorem 5.6:

$$\begin{aligned}
\big\|f - \hat{f}_n^{(\hat{L}^*)}\big\|_{L^2(\mu)} &= \big\|f - \hat{f}_n^{(L^*(n))}\big\|_{L^2(\mu)} \\
&\leq 2\|f\|_{L^\infty} \cdot \delta(L^*(n))^{1/2} + C(f,d,s) \cdot n^{-s/(2s+d)} \\
&\leq 2\|f\|_{L^\infty} \cdot (C_0 n^{-2s/(2s+d)})^{1/2} + C(f,d,s) \cdot n^{-s/(2s+d)} \\
&= O(n^{-s/(2s+d)}).
\end{aligned}$$

□

5.3 Entropy characterisation of reconstruction

Definition 5.9 (Fibre mixing)**.** Fix $\varepsilon > 0$. An event $S^* \in \mathcal{B}$ is (c,ε)*-mixing across fibres* if there exists $c > 0$ such that

$$\mu_z(S^*)\,\mu_z\big((S^*)^c\big) \;\geq\; c \cdot p_z \quad \text{for } \nu_L\text{-a.e. } \varepsilon\text{-large fibre } z.$$

Ergodicity of T is expected to force this condition (see Section 8), but the general derivability question remains open; see the open problems below. This condition is independent of (US) and is not a standing assumption here; it is used only in Corollary 5.12 below.

Lemma 5.10 (Positive-fraction balance). *Let $S^* \in \mathcal{B}$ be an ε-near-maximizer of $\delta(L)$, i.e. $\mathbb{E}[\mathrm{Var}(\mathbb{1}_{S^*} \mid \mathcal{O}_h^{(L)})] \geq \delta(L) - \varepsilon$. Define the* large-fibre contribution

$$\kappa \;:=\; \int_{\{p_z \geq \varepsilon\}} p_z\, \mu_z(S^*)\, \mu_z\big((S^*)^c\big)\, d\nu_L(z).$$

For any $\eta \in (0, \frac{1}{2})$ with $\kappa > \eta$, the set

$$G(\eta) \;:=\; \big\{z : p_z \geq \varepsilon,\ \mu_z(S^*)\, \mu_z\big((S^*)^c\big) \geq \eta\big\}$$

of ε-large balanced fibres satisfies

$$\nu_L\big(G(\eta)\big) \;\geq\; 4(\kappa - \eta).$$

No dynamical hypothesis is required.

Proof. Partition the ε-large fibres into monochromatic fibres $\mathcal{M} = \{z : p_z \geq \varepsilon,\ \mu_z(S^*)\mu_z((S^*)^c) < \eta\}$ and balanced fibres $G(\eta)$. On $\mathcal{M}$: $\mu_z(S^*)\mu_z((S^*)^c) < \eta$, so

$$\int_{\mathcal{M}} p_z\, \mu_z(S^*)\mu_z\big((S^*)^c\big)\, d\nu_L \;\leq\; \eta \int_{\mathcal{M}} p_z\, d\nu_L \;\leq\; \eta.$$

Hence $\int_{G(\eta)} p_z \mu_z(S^*)\mu_z((S^*)^c)\, d\nu_L \geq \kappa - \eta$. On $G(\eta)$: $\mu_z(S^*)\mu_z((S^*)^c) \leq \frac{1}{4}$ pointwise, so

$$\kappa - \eta \;\leq\; \int_{G(\eta)} p_z\, \mu_z(S^*)\mu_z\big((S^*)^c\big)\, d\nu_L \;\leq\; \tfrac{1}{4} \int_{G(\eta)} p_z\, d\nu_L \;\leq\; \tfrac{1}{4}\, \nu_L\big(G(\eta)\big). \qquad \square$$

Lemma 5.11 (Fibre mixing bound). *Let S^* be a $(\delta(L) - \varepsilon)$-near-maximizer of* $\mathrm{E}[\mathrm{Var}(\mathbb{1}_{\cdot} \mid \mathcal{O}_h^{(L)})]$ *that is (c, ε)-mixing across fibres (Definition 5.9). Then*

$$(\mu \otimes \mu)(R_L) \;\leq\; \frac{1}{c}\, \delta(L) + C(c)\, \varepsilon, \qquad C(c) := \frac{1+c}{c}.$$

Proof. Write $(\mu \otimes \mu)(R_L) = \int p_z^2\, d\nu_L(z)$ and split into ε-large and small-fibre parts.

Large fibres ($p_z \geq \varepsilon$). The (c, ε)-mixing condition gives $\mu_z(S^*)\mu_z((S^*)^c) \geq c\, p_z$ on ε-large fibres, so:

$$c \int_{\{p_z \geq \varepsilon\}} p_z^2\, d\nu_L(z) \;\leq\; \int p_z\, \mu_z(S^*)\mu_z\big((S^*)^c\big)\, d\nu_L(z) \;=\; \mathrm{E}\big[\mathrm{Var}(\mathbb{1}_{S^*} \mid \mathcal{O}_h^{(L)})\big] \;\leq\; \delta(L).$$

Hence $\int_{\{p_z \geq \varepsilon\}} p_z^2\, d\nu_L(z) \leq \frac{1}{c}\, \delta(L)$.

Small fibres ($p_z < \varepsilon$). Each such fibre satisfies $p_z^2 \leq \varepsilon\, p_z$, so

$$\int_{\{p_z < \varepsilon\}} p_z^2\, d\nu_L(z) \;\leq\; \varepsilon \int p_z\, d\nu_L(z) = \varepsilon.$$

Combining. Since $C(c) = \frac{1+c}{c} \geq 1$ we have $\varepsilon \leq C(c)\, \varepsilon$, and therefore

$$(\mu \otimes \mu)(R_L) \;\leq\; \frac{1}{c}\, \delta(L) + C(c)\, \varepsilon. \qquad \square$$

The collision probability (**??**) defines a natural information-theoretic quantity:

$$H_2(\nu_L) \;:=\; -\log\big((\mu \otimes \mu)(R_L)\big) \;=\; -\log \int p_z^2\, d\nu_L(z). \tag{4}$$

This is the *Rényi*-2 *(collision) entropy* of the delay-vector distribution.

Corollary 5.12 (Entropy characterisation of reconstruction)**.** *Under the fibre mixing condition of Definition 5.9:*

$$\delta(L) \to 0 \qquad \Longleftrightarrow \qquad H_2(\nu_L) \to +\infty.$$

Equivalently, reconstruction occurs if and only if the probability that two independently drawn delay vectors coincide vanishes as $L \to \infty$. Thus reconstruction is equivalent to the vanishing of collision probability $(\mu \otimes \mu)(R_L)$ introduced in Section 2.4.

Proof. Corollary 3.2 gives $\delta(L) \le \frac{1}{2}(\mu \otimes \mu)(R_L)$. For the reverse, Lemma 5.11 with a $\delta(L)$-near-maximizer S^* satisfying the (c, ε)-mixing condition gives $(\mu \otimes \mu)(R_L) \le \frac{1}{c}\,\delta(L) + C(c)\,\varepsilon$, so $\delta(L) \to 0 \iff (\mu \otimes \mu)(R_L) \to 0$. Since $(\mu \otimes \mu)(R_L) = e^{-H_2(\nu_L)}$, this is equivalent to $H_2(\nu_L) \to +\infty$. $\square$

Remark 5.13 (Balance mechanism)**.** Lemma 5.10 guarantees that a positive ν_L-fraction of ε-large fibres are η-balanced without any dynamical hypothesis. The fibre mixing condition is the upgrade from positive-fraction to ν_L-a.e. balance; whether ergodicity supplies it is the main open question (Section 8).

Remark 5.14 (Empirical entropy witness)**.** The empirical analogue of $H_2(\nu_L)$ is

$$\hat{H}_2\left(\nu_L^{(n)}\right) \;:=\; -\log\left(\frac{1}{n^2}\sum_{i,j=1}^{n} \mathbb{1}\left[\Phi_h^{(L)}(x_i) = \Phi_h^{(L)}(x_j)\right]\right),$$

the negative log of the empirical collision probability. This is the third computable witness for reconstruction. It is *computationally simpler* than $\hat{\delta}(L, n)$: it requires no optimisation over sets, only histogram counts over delay vectors. Under fibre mixing, Corollary 5.12 shows that $\hat{L}^*$ could equivalently be defined as the first L at which $\hat{H}_2(\nu_L^{(n)})$ exceeds a threshold calibrated to n.

6 The Dynamics Theorem

6.1 Lipschitz continuity of the delay map

Lemma 6.1 (Upper Lipschitz bound)**.** *Assume $T \in \mathrm{Diff}^r(X)$, $h \in C^r(X)$, $r \ge 1$. Then $\Phi_h^{(L)}$ is Lipschitz:*

$$|\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x')| \;\le\; C_L \cdot d_X(x, x') \quad \forall\, x, x' \in X,$$

where $C_L := \|Dh\|_{L^\infty} \cdot \left(\sum_{j=0}^{L} \|DT^j\|_{L^\infty}^2\right)^{1/2}$.

Proof. Component j of $\Phi_h^{(L)}$ is $h \circ T^j$. By the chain rule and compactness of X: $|h(T^j x) - h(T^j x')| \le \|Dh\|_{L^\infty} \cdot \|DT^j\|_{L^\infty} \cdot d_X(x, x')$. Taking Euclidean norm over $j = 0, \ldots, L$ gives the result. $\square$

Lemma 6.2 (Local lower Lipschitz bound)**.** *Under (G1)–(G2) with $L \ge d-1$ (Takens regime), $T \in C^r$, $r \ge 2$: for μ-a.e. $x \in X$, there exists a neighbourhood U_x and a constant $c_x = \sigma_{\min}(D\Phi_h^{(L)}|_x) > 0$ such that*

$$|\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x')| \;\ge\; c_x \cdot d_X(x, x') \quad \forall\, x' \in U_x.$$

Proof. By Lemma 5.2, $D\Phi_h^{(L)}|_x$ has full rank d for μ-a.e. x when $L \ge d-1$, so $\sigma_{\min}(D\Phi_h^{(L)}|_x) > 0$ μ-a.e. The lower bound on U_x follows from the inverse function theorem applied to the immersion $\Phi_h^{(L)}$ at x. $\square$

Remark 6.3 (Gap between local and global lower bound)**.** Lemma 6.2 gives a *local* lower bound $c_x > 0$ varying over X. (G1)–(G2) guarantee $\sigma_{\min} > 0$ μ-*a.e.*, but not on all of X. Condition (US) imposes $c := \inf_X \sigma_{\min}(D\Phi_h^{(L)}|_x) > 0$, which is strictly stronger. The gap is genuine: there exist pairs (T, h) satisfying (G1)–(G2) but violating (US) on a μ-null set.

6.2 Uniform separation

The one-step predictive kernel for deterministic T is the Dirac family $\Pi_h^{(L)}(x, \cdot) = \delta_{\Phi_h^{(L)}(x)}$. The TV separation pseudometric is therefore

$$d_L(x, x') \;:=\; \|\Pi_h^{(L)}(x, \cdot) - \Pi_h^{(L)}(x', \cdot)\|_{\mathrm{TV}} \;=\; \mathbb{1}\big[\Phi_h^{(L)}(x) \neq \Phi_h^{(L)}(x')\big] \;\in\; \{0, 1\}.$$

Separation is binary: $d_L(x, x') = 1$ if and only if x and x' are distinguished by $\mathcal{O}_h^{(L)}$.

Theorem 6.4 (Uniform separation under (US))**.** *Assume (G1)–(G2), (US), $T \in C^r$, $r \geq 2$, $L \geq d-1$. Then $\Phi_h^{(L)}$ is bi-Lipschitz on X:*

$$c \cdot d_X(x, x') \;\leq\; |\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x')| \;\leq\; C_L \cdot d_X(x, x') \quad \forall\, x, x' \in X.$$

For any estimator $\hat{\Gamma}_h^{(n)}$ with uniform error $|\hat{\Gamma}_h^{(n)}(x) - \Phi_h^{(L)}(x)| \leq \varepsilon_n$ uniformly in x, the empirical separation $\widehat{d}_L(x, x') := |\hat{\Gamma}_h^{(n)}(x) - \hat{\Gamma}_h^{(n)}(x')|$ satisfies

$$\widehat{d}_L(x, x') \;\geq\; c \cdot d_X(x, x') - 2\varepsilon_n.$$

In particular, if $d_L(x, x') = 1$ (i.e. $\Phi_h^{(L)}(x) \neq \Phi_h^{(L)}(x')$), then $\widehat{d}_L(x, x') > 0$ for all n large enough that $\varepsilon_n < c \cdot d_X(x, x')/2$.

Proof. Upper bound: Lemma 6.1. Lower bound: (US) directly gives $|\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x')| \geq c \cdot d_X(x, x')$ for all $x, x' \in X$. For the empirical bound: by the triangle inequality, $|\hat{\Gamma}_h^{(n)}(x) - \hat{\Gamma}_h^{(n)}(x')| \geq |\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x')| - 2\varepsilon_n \geq c \cdot d_X(x, x') - 2\varepsilon_n$. If $\Phi_h^{(L)}(x) \neq \Phi_h^{(L)}(x')$ then by bi-Lipschitz $d_X(x, x') \geq |\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x')|/C_L > 0$, so the right side is eventually positive. □

Theorem 6.5 (Dynamics Theorem)**.** *Assume (G1)–(G2), (US), $T \in C^r$, $r \geq 2$, μ smooth positive density on compact d-manifold X, $L \geq d-1$. Let $\hat{\Gamma}_h^{(n)}$ be the empirical delay map estimator with uniform error $\varepsilon_n = C'' n^{-\beta/(2\beta+d)}$ (Proposition 4.6). Then:*

- *(a)* (Convergence) $\|\hat{\Gamma}_h^{(n)} - \Phi_h^{(L)}\|_{L^\infty} \to 0$ *at rate* $n^{-\beta/(2\beta+d)}$.
- *(b)* (Separation) *For $\mu \otimes \mu$-a.e. (x, x') with $x \neq x'$: $\widehat{d}_L(x, x') > 0$ for all large n.*
- *(c)* (Scope) *Condition (US) is not a consequence of (G1)–(G2). Without (US), the kernel witness $\widehat{d}_L$ can vanish on a positive-measure set of pairs even when $d_L = 1$ and reconstruction holds.*

Proof. (a) Proposition 4.6 with $d_{\mathrm{eff}}(L) = d$ gives $\varepsilon_n \to 0$ at rate $n^{-\beta/(2\beta+d)}$.

(b) Since μ is non-atomic (smooth positive density), $\mu \otimes \mu$ almost every pair satisfies $d_X(x, x') > 0$. For such pairs $d_L(x, x') = 1$ (reconstruction and injectivity of $\Phi_h^{(L)}$), and by Theorem 6.4, $\widehat{d}_L(x, x') \geq c \cdot d_X(x, x') - 2\varepsilon_n$. For fixed $x \neq x'$, $c \cdot d_X(x, x') > 0$ and $\varepsilon_n \to 0$, so the right side is eventually positive.

(c) The proof of Theorem 6.4 uses (US) in the lower bound. Without it, there exist points x where $\sigma_{\min}(D\Phi_h^{(L)}|_x) \to 0$ along sequences $x'_n \to x$, and $|\Phi_h^{(L)}(x'_n) - \Phi_h^{(L)}(x)|/d_X(x, x'_n) \to 0$. Any estimator inherits this degeneracy. □

7 The Conjunction Theorem

This section formalises the complete reconstruction diagnostic and proves that both witnesses certify the same event when reconstruction and (US) hold.

7.1 The honest bridge

The delay vector $\Phi_h^{(L)}(x) = (h(x), \ldots, h(T^L x))$ is a trajectory of the Markov chain on h-space with one-step kernel Π_h. The σ-algebra $\mathcal{O}_h^{(L)} = \sigma(\Phi_h^{(L)})$ is exactly the information generated by L steps of this Markov chain starting from x. This gives an algebraic identity:

$$\sigma\big(\Phi_h^{(L)}\big) \;=\; \mathcal{O}_h^{(L)},$$

which is not an approximation but a definition. Consequently, the σ-algebra separation event and the metric separation event are identical:

Lemma 7.1 (Algebra separation equals metric separation)**.** *For deterministic T and $\mu \otimes \mu$-a.e. (x, x'):*

$$d_L(x, x') = 1 \iff x \text{ and } x' \text{ are separated by } \mathcal{O}_h^{(L)}.$$

These are the same event, not merely correlated.

Proof. For deterministic T: $\Pi_h^{(L)}(x, \cdot\,) = \delta_{\Phi_h^{(L)}(x)}$, so $d_L(x, x') = \|\delta_{\Phi_h^{(L)}(x)} - \delta_{\Phi_h^{(L)}(x')}\|_{\mathrm{TV}} = \mathbb{1}[\Phi_h^{(L)}(x) \neq \Phi_h^{(L)}(x')]$. And x is separated from x' by $\mathcal{O}_h^{(L)} = \sigma(\Phi_h^{(L)})$ if and only if $\exists E \in \mathcal{O}_h^{(L)}$ with $\mathbb{1}_E(x) \neq \mathbb{1}_E(x')$, i.e., if and only if $\Phi_h^{(L)}(x) \neq \Phi_h^{(L)}(x')$ (since $\mathcal{O}_h^{(L)}$ is generated by the level sets of $\Phi_h^{(L)}$). The two conditions are identical. $\square$

Under reconstruction ($\delta(L) = 0$, equivalently $\Phi_h^{(L)}$ injective μ-a.e.) and (US), injectivity holds everywhere, so $d_L(x, x') = 1$ for $\mu \otimes \mu$-a.e. $x \neq x'$.

7.2 The joint diagnostic

Definition 7.2 (Joint reconstruction diagnostic)**.** Fix tolerances $\tau_n, \eta_n > 0$. Define

$$\mathrm{Reconstruct}(L, n) \;:=\; \big\{\hat{\delta}(L, n) \leq \tau_n\big\} \;\wedge\; \big\{\widehat{d}_L(x, x') > \eta_n \text{ for } \mu \otimes \mu\text{-most pairs}\big\}.$$

Theorem 7.3 (Conjunction Theorem)**.** *Assume the hypotheses of Theorem 5.8 (for the algebra side) and Theorem 6.5 (for the dynamics side). Set $\tau_n = \frac{C_\lambda}{2} C' n^{-2s/(2s+d)}$ and $\eta_n = \sqrt{\varepsilon_n}$ where $\varepsilon_n = C' n^{-2s/(2s+d)}$. Then:*

- *(a)* (Conjunction fires under reconstruction + (US)). *Under $\delta(L) = 0$ for $L \geq L^*(n)$ and (US):*

$$P\big(\mathrm{Reconstruct}(\hat{L}^*, n)\big) \;\to\; 1 \quad \textit{exponentially in } n.$$

- *(b)* (Algebra witness fails when reconstruction fails). *If $\delta(L) > 0$ for all L: $P(\hat{\delta}(L, n) \leq \tau_n) \to 0$ for every fixed L, and $\hat{L}^* \to \infty$.*
- *(c)* (Kernel witness fails when (US) fails). *If $\sigma_{\min}(D\Phi_h^{(L)}|_x) \to 0$ on a positive-measure set S: $P(\widehat{d}_L > \eta_n$ for $\mu \otimes \mu$-most pairs$) \to 0$ even when $\hat{\delta}(L, n) \to 0$.*

In particular: the conjunction is not achievable when either hypothesis fails.

Proof. Part (a). By Theorem 5.6, $P(\hat{L}^* = L^*(n)) \to 1$ exponentially. On $\{\hat{L}^* = L^*(n)\}$: (i) $\hat{\delta}(\hat{L}^*, n) \le \tau_n$ whp by Theorem 4.7 and the definition of $L^*(n)$ ($\delta(L^*(n)) = 0$ under reconstruction). (ii) By Theorem 6.5(b), $\widehat{d}_{\hat{L}^*}(x, x') > 0$ for $\mu \otimes \mu$-a.e. (x, x') with $x \neq x'$, eventually. The set $\{d_X(x, x') \le (2\varepsilon_n + \eta_n)/c\}$ has $\mu \otimes \mu$-measure $\to 0$ as $\eta_n/c \to 0$, so $\widehat{d}_{\hat{L}^*} > \eta_n$ for $\mu \otimes \mu$-most pairs.

Part (b). If $\delta(L) > 0$ for all L, then by Theorem 4.7, $\hat{\delta}(L, n) \ge \delta(L) - \varepsilon_n \ge \delta(L)/2 > \tau_n$ with high probability for all fixed L, so the first condition of Reconstruct never fires.

Part (c). If $S = \{x : \sigma_{\min}(D\Phi_h^{(L)}|_x) = 0\}$ has $\mu(S) > 0$, then for $x \in S$ there exist sequences $x'_k \to x$ with $|\Phi_h^{(L)}(x'_k) - \Phi_h^{(L)}(x)|/d_X(x'_k, x) \to 0$. Any consistent estimator satisfies $\widehat{d}_L(x, x'_k) \le |\Phi_h^{(L)}(x'_k) - \Phi_h^{(L)}(x)| + 2\varepsilon_n \to 0$. The set of pairs where $\widehat{d}_L \le \eta_n$ thus has positive $\mu \otimes \mu$-measure, and the kernel condition fails. □

8 Discussion

Polynomial mixing. The Algebra Theorem (Theorem 5.8) and the Elbow Location Theorem (Theorem 5.6) require exponential mixing with resolvability constant $C_\lambda > 4$. Under polynomial mixing $\delta(L) \lesssim L^{-\alpha}$, the elbow drop at $L^*(n)$ satisfies $\Delta(L^*(n)) \sim \varepsilon_n/L^*(n) \to 0$ relative to ε_n: the elbow and the noise floor merge asymptotically and cannot be separated. A different stopping criterion (e.g. integrated elbow, or a bandwidth-adaptive rule) is needed.

The $d > 2s$ hypothesis. Proposition 4.2 requires $d > 2s$ for the Rademacher concentration to be useful at rate ε_n. When $d \le 2s$, localised Rademacher complexity (Bartlett–Bousquet–Mendelson [1]) gives adaptive concentration without this condition but requires additional work.

Uniform separation from first principles. Condition (US) is a hypothesis in Theorems 6.5 and 7.3. The natural conjecture is that T Anosov with h generic implies (US) via the uniform hyperbolicity constants. This would close the gap between (G1)–(G2) and (US) in the hyperbolic category.

Stochastic T. For stochastic dynamics, $\Pi_h^{(L)}(x, \cdot)$ is not a Dirac delta. The TV separation d_L is no longer binary, and Hölder continuity of $x \mapsto \Pi_h^{(L)}(x, \cdot)$ in TV is both necessary and non-trivial. The four-link chain of Proposition 4.6 adapts, but the Conjunction Theorem's proof structure changes substantially.

Sharp rates for the data-driven lag. Theorem 5.6 shows $P(\hat{L}^* = L^*(n)) \to 1$ but does not bound the fluctuations of $\hat{L}^*$ around $L^*(n)$ on the event $\{\hat{L}^* \neq L^*(n)\}$. A Berry–Esseen-type result for $\hat{L}^* - L^*(n)$ would sharpen the stopping rule.

Concentration of the entropy witness. The empirical collision entropy $\hat{H}_2(\nu_L^{(n)})$ is a computationally simpler stopping criterion (Remark 5.14). Asymptotic equivalence with $\hat{\delta}(L, n)$ is established under fibre mixing (Corollary 5.12), but a finite-n concentration result for $\hat{H}_2(\nu_L^{(n)})$ — comparable to Proposition 4.2 for $\hat{\delta}(L, n)$ — is not proved here. Such a result requires only McDiarmid-type bounds on the U-statistic $\frac{1}{n^2}\sum_{i,j} \mathbb{1}[\Phi_h^{(L)}(x_i) = \Phi_h^{(L)}(x_j)]$, with no new structural assumptions beyond those already made.

Derivability of fibre mixing (the main conceptual open problem). The central open question is whether the fibre mixing condition (Definition 5.9) is derivable from dynamical hypotheses, or whether it is an irreducible admissibility condition analogous to CE in [19].

Lemma 5.10 settles Step A: a positive ν_L-fraction of ε-large fibres are η-balanced, with an explicit lower bound depending only on the large-fibre contribution κ and not on any dynamical hypothesis.

The open question is Step B: what is the weakest dynamical condition that upgrades positive-fraction balance to ν_L-a.e. balance? Ergodicity is the natural candidate: in an ergodic system, a near-maximizer of $\delta(L)$ must cross fibres rather than align with them, since aligned events are $\mathcal{O}_h^{(L)}$-measurable and thus carry zero approximation error. The informal argument is clear, but formalizing the connection between ergodicity and the a.e. balance condition is the isolated open gap. If ergodicity suffices, the fibre mixing condition is derivable and the structural analogy with CE breaks; if it does not, the analogy is complete.

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